Real Numbers: Mini-Lessons

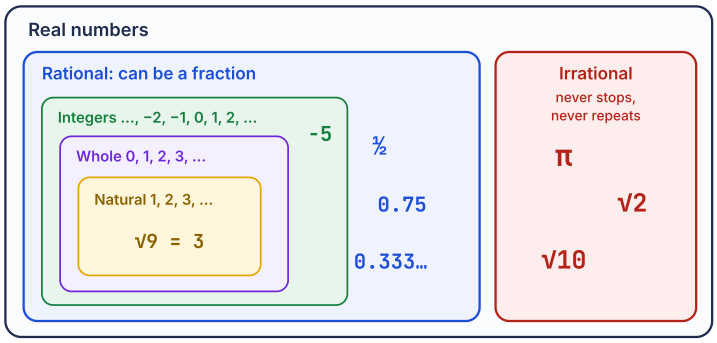
1.1 Rational or irrational? NC.8.NS.1

A number is **rational** if it can be written as a **fraction** of two integers. Its decimal **stops** or **repeats**. A number is **irrational** if its decimal **never stops and never repeats**, like π or √2.

Terminating decimal
a decimal that stops: 0.75

Repeating decimal
a decimal with digits that repeat forever: 0.333...

Perfect square
a whole number times itself: 1, 4, 9, 16, 25, 36, 49, ...



Square roots: √(perfect square) is rational (√49 = 7). √(not a perfect square) is irrational (√10).
Say it this way: "___ is rational because it can be written as a fraction."

Watch out: A long decimal is not always irrational. If it **repeats**, it is rational.

Walk through: `Is √49 rational or irrational?`

1. Is 49 a perfect square?
   Yes: 7 × 7 = 49 → 7 × 7 = 49
2. So what is √49?
   7 → √49 = 7
3. Can 7 be written as a fraction?
   Yes: 7 = 7/1, so √49 is rational → 7 = 7/1 → rational

Check your understanding

1 Score 3 NC.8.NS.1
Is `0.333…` rational or irrational?
(A) **Rational** (B) Irrational

2 Score 3 NC.8.NS.1
Is `√10` rational or irrational?
(A) **Irrational** (B) Rational

3 Score 3 NC.8.NS.1

Which number is **irrational**?

(A) π (B) −5

(C) 0.75 (D) √16

1.2 Repeating decimals to fractions (the 9's rule) NC.8.NS.1

To change a repeating decimal into a fraction: the **repeating digits go on top**. The **bottom is all 9s**, one 9 for each repeating digit. Then **simplify**.

Bar notation

a bar over the digits that repeat: $0.\overline{3}$ = 0.333...

Simplify

divide the top and bottom by the same number

$0.272727\ldots$ $0.\overline{27}$

$$\frac{27}{99} \div 9 = \frac{3}{11}$$

2 repeating digits → two 9s

Decimal	Repeating digits	Fraction	Simplified
$0.\overline{3}$	1 digit	3/9	1/3
$0.\overline{27}$	2 digits	27/99	3/11
$0.\overline{125}$	3 digits	125/999	125/999

Watch out: Count the 9s carefully: one 9 for **each** repeating digit. And always simplify.

Walk through: Write 0.363636… as a fraction

1. Which digits repeat?
   36 → $0.\overline{36}$
2. What is the bottom of the fraction?
   99 (two 9s) → 36/99
3. Simplify 36/99. What number divides both?
   9: 36/99 = 4/11 → 36/99 = 4/11

Check your understanding

1 Score 3 NC.8.NS.1

Write $0.\overline{5}$ as a fraction.

(A) 5/10 (B) **5/9** (C) 1/5 (D) 5/99

2 Score 3 NC.8.NS.1

Write $0.\overline{12}$ as a fraction in simplest form.

(A) 12/9 (B) **4/33** (C) 3/25 (D) 12/99

1.3 Square roots and cube roots NC.8.EE.2

A **square root** undoes squaring: $\sqrt{25} = 5$ because $5 \times 5 = 25$. A **cube root** undoes cubing: $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$.

Squared (^{2})

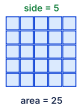
a number times itself: $6^2 = 6 \times 6 = 36$

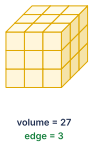
Cubed (^{3})

a number times itself 3 times: $2^3 = 2 \times 2 \times 2 = 8$

$\sqrt{}$ and $\sqrt[3]{}$

square root and cube root: they go backward





n	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	125	216	343	512	729	1000

Watch out: $\sqrt{}$ does **not** mean "divide by 2", and $\sqrt[3]{}$ does **not** mean "divide by 3". $\sqrt{16} = 4$, not 8.

Walk through: Find $\sqrt[3]{125}$

1. What does $\sqrt[3]{125}$ ask?
   What number × itself × itself = 125? → ? × ? × ? = 125
2. Try 5. What is 5 × 5 × 5?
   125 → 5 × 5 × 5 = 125
3. So $\sqrt[3]{125}$ = ?
   5 → $\sqrt[3]{125} = 5$

Check your understanding

1 Score 3 NC.8.EE.2

Find $\sqrt{81}$.

Answer: **9**

2 Score 3 NC.8.EE.2

Find $\sqrt[3]{8}$.

Answer: **2**

3 Score 3 NC.8.NS.1

Which number is a **perfect square**?

(A) 50 (B) 18

(C) 27 (D) **36**

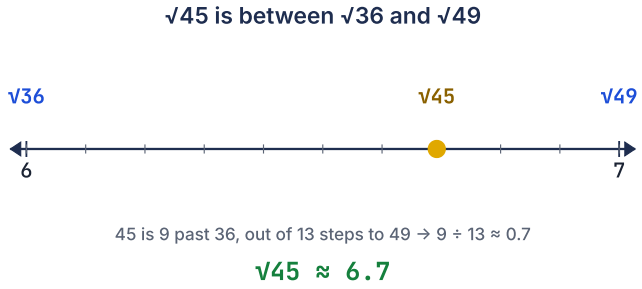
1.4 Estimate roots and order numbers NC.8.NS.2

To estimate a square root that is not a perfect square, find the **perfect squares just below and just above**. The root is **between** their roots. To compare numbers, write them all as **decimals** first.

Estimate
a close guess you can explain

Between
√45 is between 6 and 7 because 36 < 45 < 49

Least to greatest
smallest first, biggest last



4-step estimate of √45: 36 < 45 < 49 → (45 − 36) ÷ (49 − 36) = 9 ÷ 13 ≈ 0.7 → √45 ≈ 6.7

Watch out: Line up the decimals before you compare. 3.5 is bigger than 3.375 even though "375" looks long.

Walk through: `Estimate √20 to the nearest tenth`

1. Which perfect squares are just below and just above 20?
   16 and 25 → 16 < 20 < 25
2. So √20 is between...
   4 and 5 → 4 < √20 < 5
3. Find the tenths: (20 − 16) ÷ (25 − 16) = ?
   4 ÷ 9 ≈ 0.4 → √20 ≈ 4 + 0.4 = 4.4

Check your understanding

1 Score 3 NC.8.NS.2

$\sqrt{30}$ is between which two whole numbers?

- **A** 5 and 6
- B 15 and 16
- C 6 and 7
- D 25 and 36

2 Score 3 NC.8.NS.2

Which is greater: $\sqrt{50}$ or 7.5?

- **A** 7.5
- B √50
- C They are equal

3 Score 3 NC.8.NS.2

Order from least to greatest: $\sqrt{5}, 2.1, 7/3$

- A √5, 2.1, 7/3
- B 2.1, 7/3, √5
- C 7/3, √5, 2.1
- **D** 2.1, √5, 7/3

SKILL BUILDERS · UNIT 1 · SCORE 3

Real Numbers: Practice Set

NC.8.NS.1–2 · NC.8.EE.2

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.NS.1

Is $0.1212\ldots$ rational or irrational?

- **(A) Rational**
- (B) Irrational

2 Score 3 NC.8.NS.1

Is $\sqrt{144}$ rational or irrational?

- **(A) Rational**
- (B) Irrational

3 Score 3 NC.8.NS.1

Is $\sqrt{8}$ rational or irrational?

- **(A) Irrational**
- (B) Rational

4 Score 3 NC.8.NS.1

Write $0.\overline{7}$ as a fraction.

- **(A) 7/9**
- (B) 7/10
- (C) 1/7
- (D) 7/99

5 Score 3 NC.8.NS.1

Write $0.\overline{45}$ as a fraction in simplest form.

- (A) 45/100
- **(B) 5/11**
- (C) 45/99
- (D) 9/20

1. $0.\overline{45} \rightarrow 45/99$
2. $45 \div 9 = 5,\ 99 \div 9 = 11$

↓ simplify

3. $= 5/11$

6 Score 3 NC.8.EE.2

Find $\sqrt{121}$.

Answer: **11**

1. $11 \times 11 = 121$
2. $\sqrt{121} = 11$

7 Score 3 NC.8.EE.2

Find $\sqrt[3]{64}$.

Answer: **4**

1. $4 \times 4 \times 4 = 64$
2. $\sqrt[3]{64} = 4$

8 Score 3 NC.8.NS.2

$\sqrt{40}$ is between which two whole numbers?

- (A) 20 and 21
- **(B) 6 and 7**
- (C) 5 and 6
- (D) 36 and 49

9 Score 3 NC.8.NS.2

Estimate $\sqrt{50}$ to the nearest tenth.

Answer: **7.1**

1. $49 < 50 < 64 \rightarrow 7 < \sqrt{50} < 8$
2. $(50 - 49) \div (64 - 49) = 1 \div 15 \approx 0.07$
3. $\sqrt{50} \approx 7.1$

10 Score 3 NC.8.NS.2

Order from least to greatest:

$\pi, \sqrt{10}, 10/3, 3.5$

(A) √10, π, 10/3, 3.5

(B) π, √10, 10/3, 3.5

(C) 3.5, 10/3, √10, π

(D) π, 10/3, √10, 3.5

1. $\pi \approx 3.14, \sqrt{10} \approx 3.16, 10/3 \approx 3.33, 3.5 = 3.50$
2. $\pi < \sqrt{10} < 10/3 < 3.5$

11 Score 3 NC.8.EE.2

Find the mistake. Sam wrote: $\sqrt{36} = 18$. What went wrong?

(A) Sam divided 36 by 2. √36 = 6 because 6 × 6 = 36.

(B) Sam is right: 18 + 18 = 36.

(C) √36 = 9 because 4 × 9 = 36.

(D) √36 = 72 because 36 × 2 = 72.

Exponents & Scientific Notation: Mini-Lessons

NC.8.EE.1, 3–4

ANSWER KEY

2.1 What an exponent means NC.8.EE.1

An **exponent** is a shortcut for multiplying the same number over and over. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Base
the number being multiplied (in 2^5, the 2)

Exponent
how many times the base is multiplied (the 5)

Expanded form
written out the long way: 2 × 2 × 2 × 2 × 2

base 2^{5} exponent

$$2 \times 2 \times 2 \times 2 \times 2 = 32$$

NOT 2 × 5 = 10

Parentheses matter: $(-3)^2 = (-3)(-3) = 9$, but $-3^2 = -(3 \times 3) = -9$.

Watch out: 2^5 is **not** 2 × 5. Write it out to check.

Walk through: Find $(-2)^3$

1. Write $(-2)^3$ in expanded form.
   $(-2)(-2)(-2) \rightarrow (-2)(-2)(-2)$
2. Multiply the first two: $(-2)(-2) = ?$
   $4 \rightarrow = 4 \cdot (-2)$
3. Now $4 \cdot (-2) = ?$
   $-8 \rightarrow = -8$

Check your understanding

1 Score 3 NC.8.EE.1

Find the value of 3^4.

Answer: **81**

2 Score 3 NC.8.EE.1

Which is the same as 5^3?

(A) 5 + 5 + 5
(B) 5 × 3
(C) 3 × 3 × 3 × 3 × 3
(D) **5 × 5 × 5**

3 Score 3 NC.8.EE.1

Find the value of $(-4)^2$.

Answer: **16**

2.2 Multiply and divide powers NC.8.EE.1

Same base: when you **multiply**, **add** the exponents. When you **divide**, **subtract** the exponents. $x^a \cdot x^b = x^{a+b}$ and $x^a \div x^b = x^{a-b}$.

Same base
the numbers or letters being raised to powers match: 2^3 and 2^4

Coefficient
the number in front: in $3x^2$, the 3

$$x^3 \cdot x^4$$

$$(x \cdot x \cdot x) \cdot (x \cdot x \cdot x \cdot x)$$

7 x's multiplied

$$x^3 \cdot x^4 = x^{3+4} = x^7$$

Multiply same base → ADD the exponents

$2^3 \times 2^4 = 2^{3+4} = 2^7$

$x^5 \div x^2 = x^{5-2} = x^3$

$(3x^2)(2x^4) = 6x^6$ (multiply the numbers, add the exponents)

Watch out: Do **not** multiply the bases. $2^3 \times 2^4$ is 2^7, not 4^7.

Walk through: Simplify $(4x^3)(2x^5)$

1. First, the numbers in front. 4 × 2 = ?
   8 → 4 × 2 = 8
2. Next, the x's: $x^3 \cdot x^5$. What do we do with the exponents?
   Add them: 3 + 5 = 8 → $x^3 \cdot x^5 = x^8$
3. Put it together.
   $8x^8$ → $(4x^3)(2x^5) = 8x^8$

Check your understanding

1 Score 3 NC.8.EE.1

Simplify: $5^2 \times 5^5$

(A) 5^{10} (B) 5^7 (C) 5^3 (D) 25^7

2 Score 3 NC.8.EE.1

Simplify: $a^7 \div a^3$

(A) a^4 (B) $a^{7/3}$ (C) a^{21} (D) a^{10}

2.3 Power of a power, zero and negative exponents NC.8.EE.1

Power of a power: multiply the exponents, $(x^3)^2 = x^6$. **Zero exponent:** any nonzero number to the 0 power is **1**. **Negative exponent:** flip it to a fraction, $2^{-3} = 1/2^3 = 1/8$.

Reciprocal
the flip of a number: the reciprocal of 5 is 1/5

Power of a power
a power raised to another power, like $(x^3)^2$

Rule	Example
$(x^a)^b = x^{a \cdot b}$	$(3^2)^3 = 3^6$
$x^0 = 1$	$9^0 = 1$
$x^{-a} = 1/x^a$	$10^{-2} = 1/100$

$2^3 = 8$
$2^2 = 4$ ÷ 2
$2^1 = 2$ ÷ 2
$2^0 = 1$ ÷ 2
$2^{-1} = 1/2$ ÷ 2
$2^{-2} = 1/4$ ÷ 2

Watch out: A negative exponent does **not** make the number negative. $2^{-3} = 1/8$, not -8.

Walk through: Write 5^{-2} as a fraction

1. The exponent is negative. What does that tell us?
   Flip it: $5^{-2} = 1/5^2 \rightarrow 5^{-2} = 1/5^2$
2. What is 5^2?
   $25 \rightarrow = 1/25$

Check your understanding

1 Score 3 NC.8.EE.1

Simplify: $(x^4)^2$

(A) x^2 (B) x^8 (C) x^6 (D) x^{16}

2 Score 3 NC.8.EE.1

Find the value of 9^0.

Answer: 1

3 Score 3 NC.8.EE.1

Find the value of 2^{-3}. (Type a fraction like 1/4.)

Answer: 1/8

2.4 Scientific notation NC.8.EE.3

Scientific notation writes a number as $a \times 10^n$, where **a is at least 1 and less than 10**. Big numbers get a **positive** exponent. Small numbers (less than 1) get a **negative** exponent.

Front number (a)

must be 1 or more, and less than 10: 4.5 ✓, 45 ×, 0.45 ×

Power of 10

the exponent counts how many places the decimal point moves



$0.00032 = 3.2 \times 10^{-4}$

front number 4.5 is between 1 and 10 ✓

Standard form	Hops	Scientific notation
45,000,000	7 left	4.5×10^7
0.00032	4 right	3.2×10^{-4}
6,020,000	6 left	6.02×10^6

Watch out: 45×10^3 is not finished: 45 is not less than 10. Fix it to 4.5×10^4.

Walk through: `Write 0.0061 in scientific notation`

1. Where should the decimal point go so the front number is between 1 and 10?
   After the 6: 6.1 → 6.1
2. How many places did the point hop?
   3 places to the right → 3 hops right
3. Positive or negative exponent?
   Negative: $6.1 \times 10^{-3} \rightarrow 0.0061 = 6.1 \times 10^{-3}$

Check your understanding

1 Score 3 NC.8.EE.3

Write `32,000` in scientific notation.

Ⓐ 3.2×10^{-4} Ⓑ 3.2×10^{3}

Ⓒ **3.2×10^{4}** Ⓓ 32×10^{3}

2 Score 3 NC.8.EE.3

Write 5×10^{3} in standard form.

Answer: **5000**

2.5 Multiply and divide in scientific notation NC.8.EE.4

Multiply: multiply the front numbers and **add** the exponents. **Divide:** divide the front numbers and **subtract** the exponents. Then check that the front number is between 1 and 10.

Adjust
if the front number is 10 or more, move the point and fix the exponent: $15 \times 10^{7} = 1.5 \times 10^{8}$

$$(8 \times 10^{7}) \div (4 \times 10^{3}) = (8 \div 4) \times 10^{7-3} = 2 \times 10^{4}$$

$$(3 \times 10^{4})(2 \times 10^{5})$$
$$= (3 \times 2) \times 10^{4+5}$$
$$= 6 \times 10^{9}$$

front numbers: multiply · tens: add exponents

Watch out: After multiplying, check the front number. 15×10^{7} must be fixed to 1.5×10^{8}.

Walk through: `Multiply (5 × 10³)(3 × 10⁴)`

1. Multiply the front numbers.
   $5 \times 3 = 15 \rightarrow 15$
2. Now the powers of 10: $10^{3} \cdot 10^{4} = ?$
   10^{7} (add 3 + 4) → 15×10^{7}
3. 15 is not less than 10. Fix it.
   $1.5 \times 10^{8} \rightarrow = 1.5 \times 10^{8}$

Check your understanding

1 Score 3 NC.8.EE.4

Multiply: $(2 \times 10^3)(4 \times 10^2)$

(A) 8×10^6

(B) 8×10^1

(C) **8×10^5**

(D) 6×10^5

2 Score 3 NC.8.EE.4

Divide: $(9 \times 10^8) \div (3 \times 10^2)$

(A) **3×10^6**

(B) 3×10^4

(C) 6×10^6

(D) 3×10^{10}

SKILL BUILDERS · UNIT 2 · SCORE 3

Exponents & Scientific Notation: Practice Set

NC.8.EE.1, 3–4

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.EE.1

Simplify: $4^3 \times 4^2$

(A) 16^5 (B) **4^5**

(C) 4^6 (D) 4^1

2 Score 3 NC.8.EE.1

Simplify: $x^7 \div x^4$

(A) $x^{7/4}$ (B) x^{11}

(C) x^{28} (D) **x^3**

3 Score 3 NC.8.EE.1

Simplify: $(2^3)^2$

(A) **2^6** (B) 4^3

(C) 2^9 (D) 2^5

4 Score 3 NC.8.EE.1

Find the value of 5^0.

Answer: **1**

5 Score 3 NC.8.EE.1

Find the value of 3^{-2}. (Type a fraction like 1/4.)

Answer: **1/9**

1. $3^{-2} = 1/3^2$
2. $= 1/9$

6 Score 3 NC.8.EE.1

Find the value of $(-3)^2$.

Answer: **9**

1. $(-3)^2 = (-3)(-3)$
2. $= 9$

7 Score 3 NC.8.EE.3

Write 0.00056 in scientific notation.

(A) 5.6×10^{-3} (B) **5.6×10^{-4}**

(C) 56×10^{-5} (D) 5.6×10^4

8 Score 3 NC.8.EE.3

Write 7.2×10^6 in standard form.

Answer: **7200000**

1. 7.2×10^6: move the point 6 places right
2. $= 7{,}200{,}000$

9 Score 3 NC.8.EE.4

Multiply: $(5 \times 10^3)(3 \times 10^4)$

- **A** 1.5×10^8
- B 1.5×10^7
- C 15×10^7
- D 8×10^7

10 Score 3 NC.8.EE.4

Divide: $(8 \times 10^7) \div (4 \times 10^3)$

- A 2×10^{10}
- **B** 2×10^4
- C 4×10^4
- D 32×10^4

11 Score 3 NC.8.EE.1

Find the mistake. Lee wrote: $x^3 \cdot x^4 = x^{12}$. What went wrong?

- **A** **Lee multiplied the exponents. Multiplying powers adds them: x^7.**
- B Lee is right.
- C It should be $2x^7$.
- D It should be x^1 because $4 - 3 = 1$.

Equations & Inequalities: Mini-Lessons

NC.8.EE.7

ANSWER KEY

3.1 One- and two-step equations NC.8.EE.7

To **solve** an equation, get **x alone**. Undo each move with its **opposite**, and do it to **both sides** so the equation stays balanced.

Variable

a letter, like x, that stands for a number

Inverse (opposite) operation

the move that undoes another: + and − undo each other, × and ÷ undo each other

Solution

the number that makes the equation true

To undo...	+ 5	− 4	× 3 (3x)	÷ 2
do this	− 5	+ 4	÷ 3	× 2

Order: undo + or − first, then × or ÷.

Watch out: Do the same move to **both sides**. A move on one side only breaks the balance.

x = 4

Check: 2(4) + 3 = 11 ✓



Walk through: `Solve 3x - 4 = 11`

1. Which move comes first?
   Add 4 to both sides → 3x = 15
2. x is multiplied by 3. What undoes that?
   Divide both sides by 3 → x = 5
3. Check: put 5 in for x. What is 3(5) − 4?
   11, so x = 5 is right → 3(5) − 4 = 15 − 4 = 11 ✓

Check your understanding

1 Score 3 NC.8.EE.7

Solve: **x + 7 = 15**

Answer: 8

1. $x + \cancel{7} = 15$ ↓ Subtract 7
   $\cancel{-7} \quad -7$
2. $x = 8$

2 Score 3 NC.8.EE.7

What is the **first** move to solve **4x + 5 = 21**?

- (A) Add 5 to both sides
- (B) Divide both sides by 4
- (C) **Subtract 5 from both sides**
- (D) Subtract 4 from both sides

3 Score 3 NC.8.EE.7

Solve: **2x + 3 = 11**

Answer: 4

1. $2x + \cancel{3} = 11$ ↓ Subtract 3
   $\cancel{-3} \quad -3$
2. $\frac{\cancel{2}x}{\cancel{2}} = \frac{8}{2}$ ↓ Divide by 2
3. $x = 4$

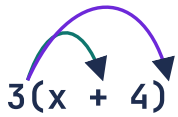
3.2 Distribute and combine like terms NC.8.EE.7

The number outside the parentheses **multiplies every term inside**. **Like terms** (x-terms with x-terms, numbers with numbers) can be added together.

Distribute
multiply the outside number by each term inside the ( )

Like terms
terms with the same variable part: 3x and 5x, or 4 and 9

Coefficient
the number in front of a variable (in 3x, the 3)



3 · x + 3 · 4

3x + 12

`4(x - 2) = 4x - 8`

`3x + 5 + 2x = 5x + 5`

Watch out: 3(x + 4) is **not** 3x + 4. The 3 multiplies the 4 too: 3x + 12.

Walk through: `Solve 2(x + 5) = 16`

1. There are parentheses. What is the first move?
   Distribute: 2 · x + 2 · 5 → `2x + 10 = 16`
2. Undo the + 10.
   Subtract 10 from both sides → `2x = 6`
3. Get x alone.
   Divide both sides by 2 → `x = 3`

Check your understanding

1 Score 3 NC.8.EE.7

Which is equal to `3(x + 4)`?

(A) 3x + 4
(B) x + 12
(C) **3x + 12**
(D) 3x + 7

2 Score 3 NC.8.EE.7

Combine like terms: `5x + 2 + 3x`

(A) $8x^2$
(B) 15x + 2
(C) 10x
(D) **8x + 2**

3 Score 3 NC.8.EE.7

Solve: `4(x - 2) = 20`

Answer: 7

1. $4(x - 2) = 20$ ↓ Distribute 4
2. $4x - 8 = 20$; $+8 \quad +8$ ↓ Add 8
3. $\frac{4x}{4} = \frac{28}{4}$ ↓ Divide by 4
4. $x = 7$

3.3 Variables on both sides NC.8.EE.7

When x is on **both sides**, move the x-terms to one side first (subtract the smaller x-term from both sides). Then solve the two-step equation that is left.

Like terms

5x and 3x are like terms, so 5x − 3x = 2x

Both sides

the left of the = sign and the right of the = sign

The 5-step plan:

1. Distribute (if there are parentheses).
2. Combine like terms on each side.
3. Move the x-terms to one side.
4. Move the numbers to the other side.
5. Divide to get x alone.

Watch out: Subtract the x-term from **both** sides, not just the side it is on.

1. $5x + 2 = 3x + 10$; $-3x \quad -3x$ ↓ Subtract 3x
2. $2x + 2 = 10$; $-2 \quad -2$ ↓ Subtract 2
3. $\frac{2x}{2} = \frac{8}{2}$ ↓ Divide by 2
4. $x = 4$

Walk through: `Solve 7x + 3 = 4x + 15`

1. x is on both sides. What move puts the x's together?
   Subtract 4x from both sides → `3x + 3 = 15`
2. Undo the + 3.
   Subtract 3 from both sides → `3x = 12`

3 Get x alone.
Divide both sides by 3 → x = 4

Check your understanding

1 Score 3 NC.8.EE.7

What is a good **first** move for $6x - 1 = 2x + 11$?

- (A) **Subtract 2x from both sides**
- (B) Divide both sides by 6
- (C) Add 2x to both sides
- (D) Subtract 6x from the right side only

2 Score 3 NC.8.EE.7

Solve: $6x - 1 = 2x + 11$

Answer: **3**

1. $6x - 1 = 2x + 11$ → Subtract 2x
   $-2x \qquad -2x$
2. $4x - 1 = 11$ → Add 1
   $+1 \qquad +1$
3. $\frac{4x}{4} = \frac{12}{4}$ → Divide by 4
4. $x = 3$

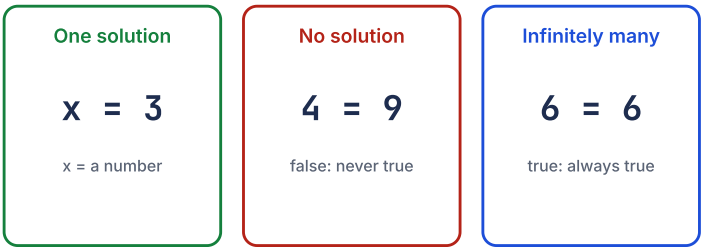
3.4 One, none, or infinitely many solutions NC.8.EE.7a

Solve as usual. If the x's cancel, look at what is left. **False** (like 4 = 9) means **no solution**. **True** (like 6 = 6) means **infinitely many solutions**. If you get x = a number, there is **one solution**.

No solution
no number makes the equation true

Infinitely many solutions
every number makes the equation true

One solution
exactly one number works, like x = 3



Look at the LAST line after the x's are gone.

`2x + 4 = 2x + 9 → 4 = 9 (false) → no solution`

`3(x + 2) = 3x + 6 → 6 = 6 (true) → infinitely many`

Watch out: If the x's cancel, you are not stuck. The last line tells you the answer.

Walk through: `How many solutions? 3(x + 2) = 3x + 6`

1. First move?
   Distribute the 3 → `3x + 6 = 3x + 6`
2. Subtract 3x from both sides. What is left?
   `6 = 6` → `6 = 6`
3. 6 = 6 is true. How many solutions?
   Infinitely many solutions → 6 = 6 is true → infinitely many

Check your understanding

1 Score 3 NC.8.EE.7a

How many solutions does **`2x + 4 = 2x + 9`** have?

- **(A) No solution**
- (B) One solution
- (C) Infinitely many solutions

2 Score 3 NC.8.EE.7a

A last line says **`5 = 5`**. How many solutions?

- **(A) Infinitely many solutions**
- (B) One solution
- (C) No solution

3 Score 3 NC.8.EE.7a

How many solutions does $4x + 1 = 2x + 7$ have?

A One solution

B No solution

C Infinitely many solutions

---

1 $4x + 1 = 2x + 7$

$-2x \quad -2x$

↓ Subtract 2x

2 $2x + 1 = 7$

$-1 \quad -1$

↓ Subtract 1

3 $\frac{2x}{2} = \frac{6}{2}$

↓ Divide by 2

4 $x = 3$

3.5 Inequalities: solve and graph NC.8.EE.7

Solve an inequality like an equation, with **one special rule**: when you **multiply or divide by a negative number, flip the sign**. Then graph it on a number line.

< and >

less than, greater than: open circle (the number is not included)

≤ and ≥

at most, at least: closed circle (the number is included)

Flip

turn the sign around: < becomes >, ≤ becomes ≥



x > 2: open circle at 2, shade right

Watch out: Only flip when you **multiply or divide by a negative**. Adding or subtracting never flips.

$-2x < 6$

$\div(-2)\ \div(-2)$

$x > -3$



Walk through: Solve 2x + 1 ≥ 7 and graph it

1. Undo the + 1.
   Subtract 1 from both sides: 2x ≥ 6 → 2x ≥ 6
2. Undo the × 2. Do we flip?
   Divide by 2. No flip: x ≥ 3 → x ≥ 3
3. Open or closed circle at 3?
   Closed: ≥ includes 3 → closed circle at 3
4. Which way do we shade?
   Right: numbers greater than 3 → shade to the right

Check your understanding

1 Score 3 NC.8.EE.7

Solve: $x + 4 > 9$

(A) x ≥ 5
(B) **x > 5**
(C) x < 5
(D) x > 13

2 Score 3 NC.8.EE.7

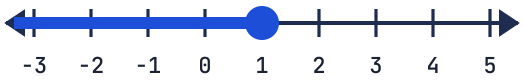
Solve: $-2x < 8$

(A) **x > −4**
(B) x < −4
(C) x < −16
(D) x > 4

3 Score 3 NC.8.EE.7

Which graph shows $x \leq 1$?

A



B



C



SKILL BUILDERS · UNIT 3 · SCORE 3

Equations & Inequalities: Practice Set

NC.8.EE.7

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.EE.7

Solve: $x - 7 = 10$

Answer: **17**

1. $x - 7 = 10$ → Add 7
   $+7 \quad +7$
2. $x = 17$

2 Score 3 NC.8.EE.7

Solve: $3x = 18$

Answer: **6**

1. $\frac{3x}{3} = \frac{18}{3}$ → Divide by 3
2. $x = 6$

3 Score 3 NC.8.EE.7

Solve: $2x + 5 = 19$

Answer: **7**

1. $2x + 5 = 19$ → Subtract 5
   $-5 \quad -5$
2. $\frac{2x}{2} = \frac{14}{2}$ → Divide by 2
3. $x = 7$

4 Score 3 NC.8.EE.7

Solve: $4x - 3 = 25$

Answer: **7**

1. $4x - 3 = 25$ → Add 3
   $+3 \quad +3$
2. $\frac{4x}{4} = \frac{28}{4}$ → Divide by 4
3. $x = 7$

5 Score 3 NC.8.EE.7

Solve: $4(x + 1) = 24$

Answer: 5

1. $4(x + 1) = 24$ ↓ Distribute 4
2. $4x + 4 = 24$, $-4 \quad -4$ ↓ Subtract 4
3. $\frac{4x}{4} = \frac{20}{4}$ ↓ Divide by 4
4. $x = 5$

6 Score 3 NC.8.EE.7

Solve: $5x - 2 = 3x + 8$

Answer: 5

1. $5x - 2 = 3x + 8$, $-3x \quad -3x$ ↓ Subtract 3x
2. $2x - 2 = 8$, $+2 \quad +2$ ↓ Add 2
3. $\frac{2x}{2} = \frac{10}{2}$ ↓ Divide by 2
4. $x = 5$

7 Score 3 NC.8.EE.7

Solve: $3(x - 2) + x = 10$

Answer: 4

1. $3(x - 2) + x = 10$ ↓ Distribute 3
2. $3x - 6 + x = 10$ ↓ Combine like terms
3. $4x - 6 = 10$, $+6 \quad +6$ ↓ Add 6
4. $\frac{4x}{4} = \frac{16}{4}$ ↓ Divide by 4
5. $x = 4$

8 Score 3 NC.8.EE.7

How many solutions does $2x + 6 = 2x + 1$ have?

(A) **No solution** (B) One solution

(C) Infinitely many solutions

1. $2x + 6 = 2x + 1$, $-2x \quad -2x$ ↓ Subtract 2x
2. $6 = 1$

1 $6 = 1$ is false → no solution

9 Score 3 NC.8.EE.7

Solve: $-3x \leq 12$

(A) $x \geq -4$ (B) $x \geq 4$

(C) $x \leq -4$ (D) $x \leq -36$

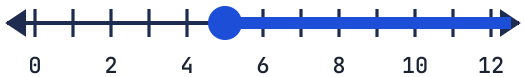
1 $-3x \leq 12$

2 $x \geq -4$ ↓ Divide by −3 and FLIP the sign

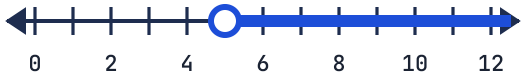
10 Score 3 NC.8.EE.7

Which graph shows $x + 4 > 9$?

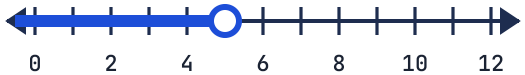
(A) 5



(B) 5



(C) 5



(D) 9



1 $x + 4 > 9$

2 $x > 5$ ↓ Subtract 4

3 open circle at 5, shade right

11 Score 3 NC.8.EE.7

Find the mistake. Jordan solved $2(x + 3) = 14$:

Line 1: $2x + 3 = 14$

Line 2: $2x = 11$

Line 3: $x = 5.5$

Where is the mistake?

A **Line 1: the 2 was not multiplied by 3. It should be 2x + 6 = 14.**

B Line 2: 14 − 3 is not 11.

C There is no mistake.

D Line 3: 11 ÷ 2 is not 5.5.

1. $2(x + 3) = 14$ ↓ Distribute 2 to BOTH terms
2. $2x + 6 = 14$ ↓ Subtract 6
   $-6 \quad -6$
3. $\frac{2x}{2} = \frac{8}{2}$ ↓ Divide by 2
4. $x = 4$

12 Score 3 NC.8.EE.7

Five more than twice a number is 17. The equation is $2x + 5 = 17$. What is the number?

Answer: **6**

1. $2x + 5 = 17$ ↓ Subtract 5
   $-5 \quad -5$
2. $\frac{2x}{2} = \frac{12}{2}$ ↓ Divide by 2
3. $x = 6$

Angle Relationships: Mini-Lessons

4.1 Angle pairs NC.8.G.5

Angles come in pairs that follow rules. **Complementary** angles add to **90°**. **Supplementary** angles add to **180°**. **Vertical** angles (across from each other where two lines cross) are **equal**.

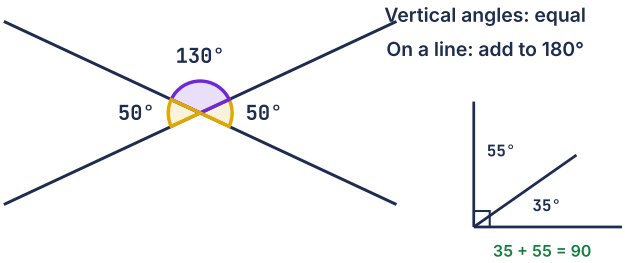
Complementary
two angles that add to 90° (a corner)

Supplementary
two angles that add to 180° (a straight line)

Vertical angles
across from each other at a crossing: equal

Adjacent
next to each other, sharing a side

Watch out: Mixing up 90 and 180?
Complementary = **C**orner = 90°.
Supplementary = **S**traight = 180°.



Walk through: `Two angles are supplementary. One is 110°. Find the other.`

1. Supplementary angles add to...
   180° → ? + 110 = 180
2. How do we find the missing angle?
   Subtract: 180 − 110 → 180 − 110 = 70
3. The other angle is...
   70° → The other angle = 70°

Check your understanding

1 Score 3 NC.8.G.5

Find the **complement** of 62°.

Answer: **28°**

1 `90 - 62 = 28°`

2 Score 3 NC.8.G.5

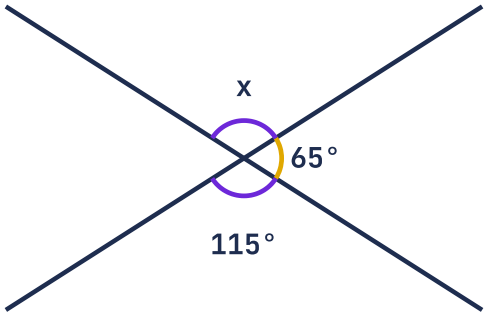
Find the **supplement** of 145°.

Answer: **35°**

1 `180 - 145 = 35°`

3 Score 3 NC.8.G.5

Two lines cross. Find **x**.



Answer: **115°**

1 `x and 115° are vertical angles → equal`

2 `x = 115°`

4.2 Parallel lines cut by a transversal NC.8.G.5

When a transversal cuts two **parallel** lines, every angle is either **equal** to a given angle or **adds with it to 180°**. Equal pairs: **corresponding, alternate interior, alternate exterior, vertical**. Same-side interior angles add to **180°**.

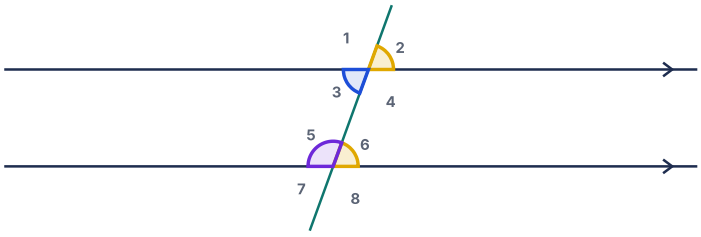
Transversal
a line that crosses two other lines

Corresponding
same corner at each crossing: equal

Alternate interior
between the lines, opposite sides: equal

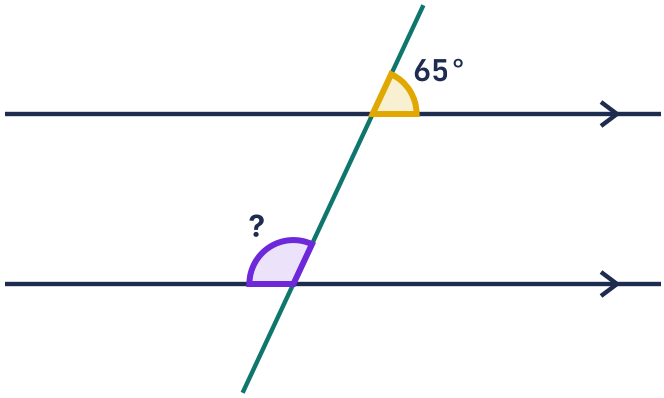
Same-side interior
between the lines, same side: add to 180°

Watch out: Look at the picture: a small-looking angle can only equal another small-looking angle.



small = 70°, big = 110°

Walk through: `Lines are parallel. ∠2 = 65°. Find ∠5.`

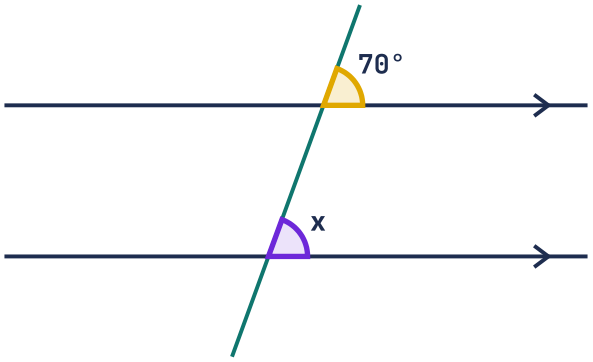


1. Is ∠5 small-looking (acute) or big-looking (obtuse)?
   Big (obtuse) → ∠2 is small, ∠5 is big
2. So ∠5 and ∠2 are...
   Supplementary: they add to 180° → ∠5 + 65° = 180°
3. Find ∠5.
   115° → ∠5 = 180 − 65 = 115°

Check your understanding

1 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

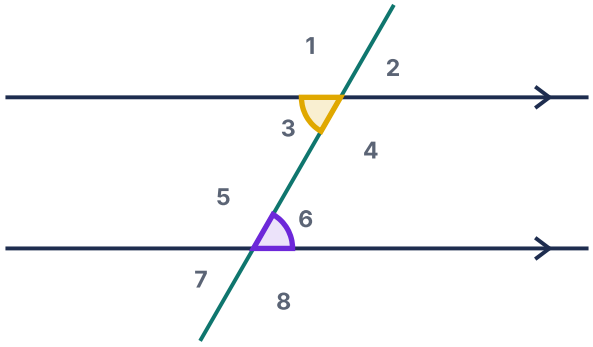


Answer: **70°**

1. `corresponding angles → equal`
2. `x = 70°`

2 Score 3 NC.8.G.5

What kind of pair are ∠3 and ∠6?



Ⓐ Corresponding angles

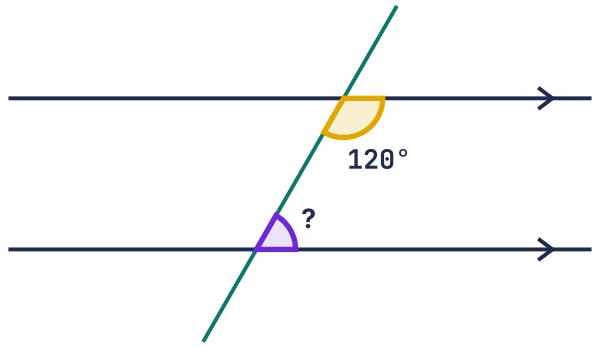
Ⓑ **Alternate interior angles**

Ⓒ Vertical angles

Ⓓ Same-side interior angles

3 Score 3 NC.8.G.5

The lines are parallel. ∠4 = 120°. Find ∠6 (a same-side interior angle).



Answer: **60°**

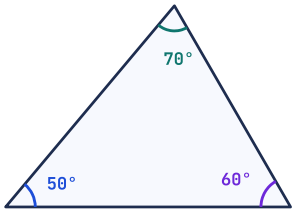
1 `180 - 120 = 60°`

4.3 Angles in a triangle NC.8.G.5

The three inside angles of **any** triangle add to **180°**. To find a missing angle: add the two you know, then subtract from 180.

Interior angle
an angle inside the triangle

Right triangle
a triangle with one 90° angle (a square corner)

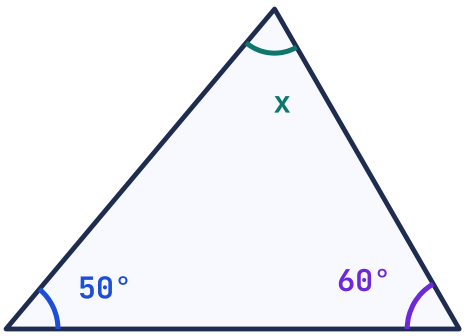


`50 + 70 + 60 = 180`



Watch out: Use 180°, not 360°. 360° is a full circle (or a four-sided shape).

Walk through: `A triangle has angles 50° and 60°. Find the third angle.`

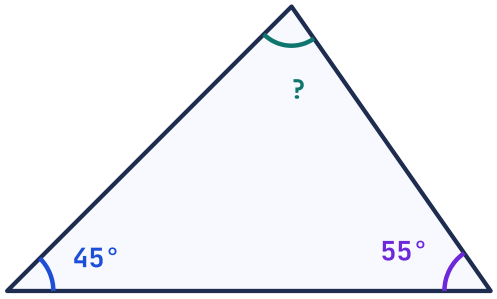


1. The three angles of a triangle add to...
   180° → 50 + 60 + x = 180
2. Add the angles you know.
   50 + 60 = 110 → 110 + x = 180
3. Find x.
   180 − 110 = 70° → x = 70°

Check your understanding

1 Score 3 NC.8.G.5

A triangle has angles 45° and 55°. Find the third angle.

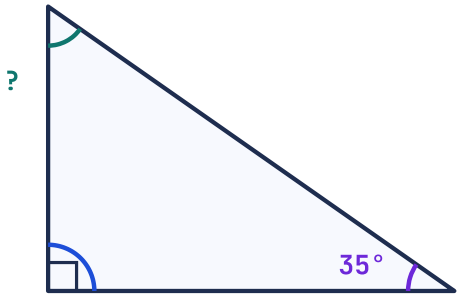


Answer: **80°**

1. `45 + 55 = 100`
2. `180 - 100 = 80°`

2 Score 3 NC.8.G.5

A right triangle has one angle of 35°. Find the other small angle.



Answer: **55°**

1. `90 + 35 = 125`
2. `180 - 125 = 55°`

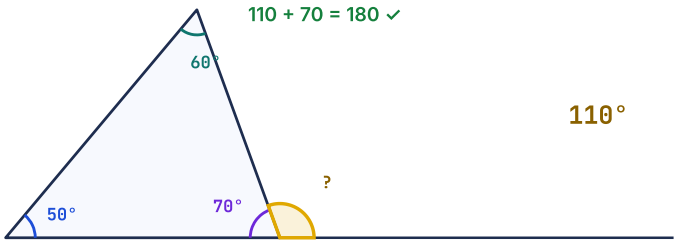
4.4 The exterior angle of a triangle NC.8.G.5

An **exterior angle** is outside the triangle, made by stretching one side. It **equals the two far inside angles added together**.

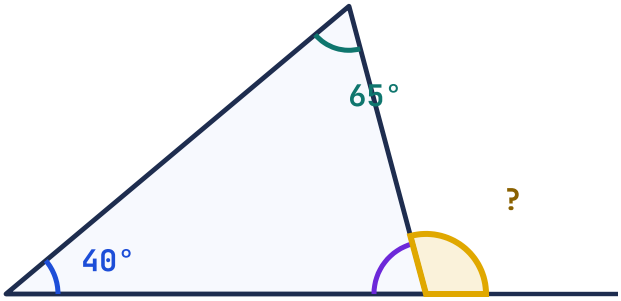
Exterior angle
the angle outside the triangle, next to one inside angle

Remote (far) interior angles
the two inside angles that do not touch the exterior angle

Watch out: Use the two **far** angles, not the angle right next to the exterior angle.



Walk through: The far inside angles are 40° and 65°. Find the exterior angle.



1. Which angles do we use?
   The two far inside angles: 40° and 65° → exterior = 40 + 65

2. Add them.
   105° → exterior = 105°

Check your understanding

1 Score 3 NC.8.G.5

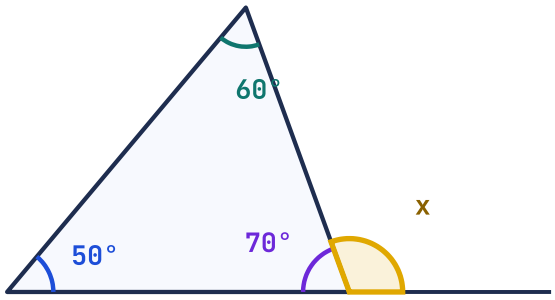
The far inside angles are 40° and 65°. Find the exterior angle.

Answer: **105°**

1. 40 + 65 = 105°

2 Score 3 NC.8.G.5

Find the exterior angle **x**.



Answer: **110°**

1. far angles: 50° and 60°
2. x = 50 + 60 = 110°

SKILL BUILDERS · UNIT 4 · SCORE 3

Angle Relationships: Practice Set

NC.8.G.5

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.5

Find the **complement** of 28°.

Answer: **62°**

1. `90 - 28 = 62°`

2 Score 3 NC.8.G.5

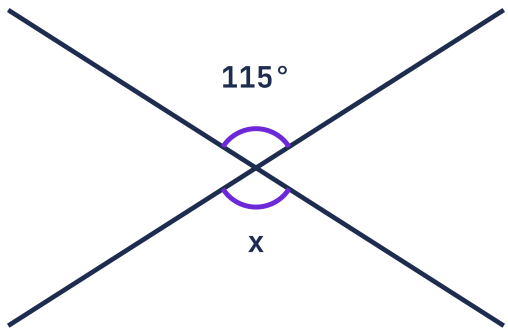
Find the **supplement** of 73°.

Answer: **107°**

1. `180 - 73 = 107°`

3 Score 3 NC.8.G.5

Two lines cross. Find **x**.

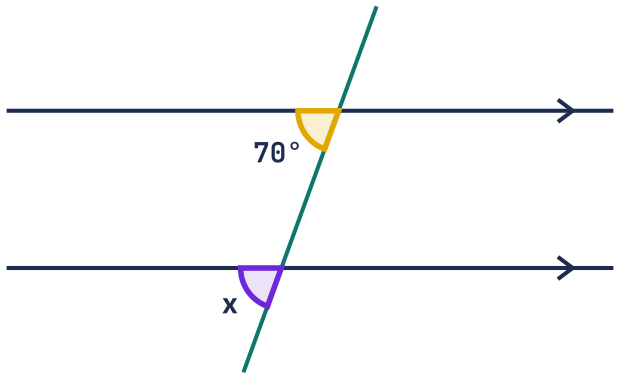


Answer: **115°**

1. `vertical angles → equal`
2. `x = 115°`

4 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

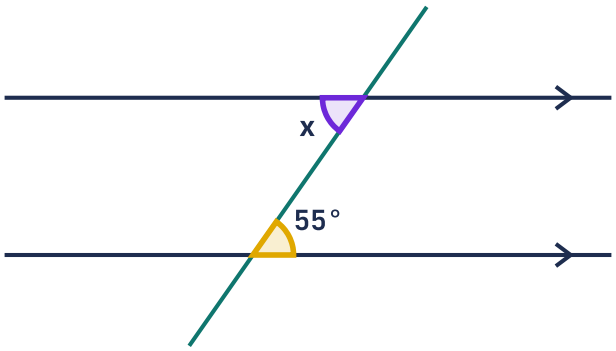


Answer: **70°**

1. `corresponding → equal`
2. `x = 70°`

5 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

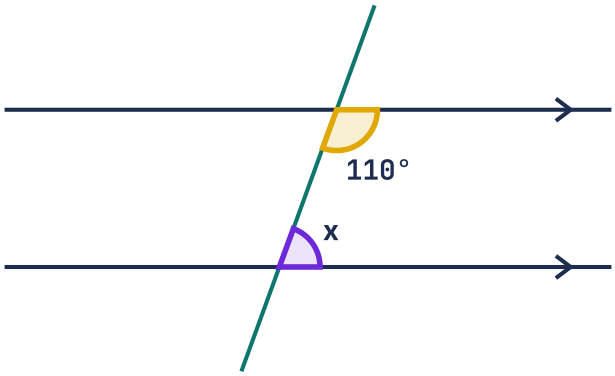


Answer: **55°**

1. alternate interior → equal
2. x = 55°

6 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

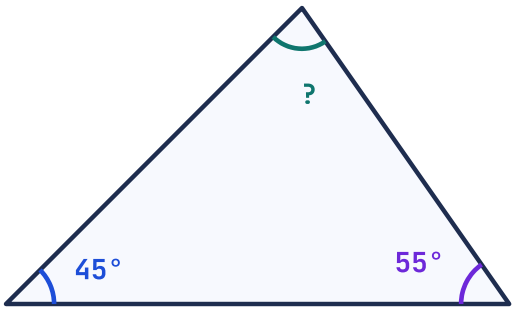


Answer: **70°**

1. same-side interior → add to 180°
2. x = 180 - 110 = 70°

7 Score 3 NC.8.G.5

A triangle has angles 45° and 55°. Find the third angle.

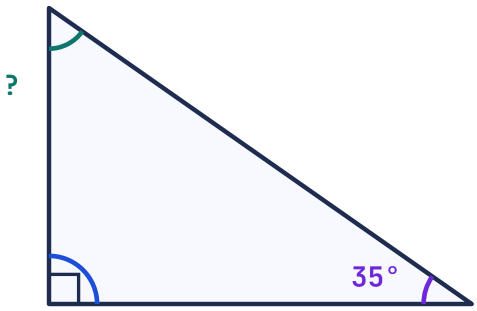


Answer: **80°**

1. 45 + 55 = 100
2. 180 - 100 = 80°

8 Score 3 NC.8.G.5

A right triangle has one angle of 35°. Find the other small angle.



Answer: **55°**

1. 90 + 35 = 125
2. 180 - 125 = 55°

9 Score 3 NC.8.G.5

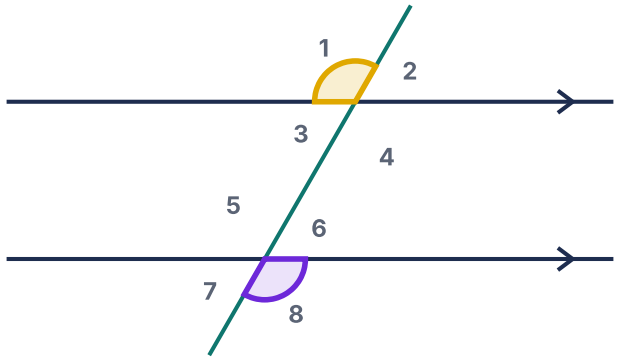
The far inside angles of a triangle are 40° and 65°. Find the exterior angle.

Answer: **105°**

1 `40 + 65 = 105°`

10 Score 3 NC.8.G.5

What kind of pair are ∠1 and ∠8?



Ⓐ Alternate interior angles

Ⓑ **Alternate exterior angles**

Ⓒ Corresponding angles

Ⓓ Same-side interior angles

11 Score 3 NC.8.G.5

Find the mistake. Ava wrote: "The supplement of 40° is 50°." What went wrong?

Ⓐ **Ava used 90. Supplementary angles add to 180, so the answer is 140°.**

Ⓑ Ava is right: 40 + 50 = 90.

Ⓒ It should be 40° because the angles are equal.

Ⓓ It should be 320° because 360 − 40 = 320.

Pythagorean Theorem: Mini-Lessons

5.1 Legs, hypotenuse and $a^2 + b^2 = c^2$ NC.8.G.6

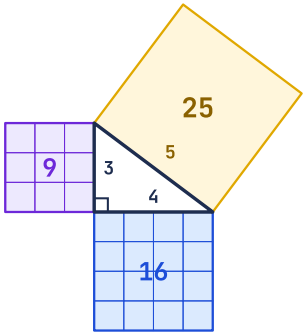
In a **right triangle**, the two shorter sides are the **legs** (a and b). The longest side, across from the right angle, is the **hypotenuse** (c). The theorem: **$a^2 + b^2 = c^2$**.

Right triangle
a triangle with a 90° angle (a square corner)

Leg
one of the two sides that make the square corner

Hypotenuse
the longest side, across from the square corner

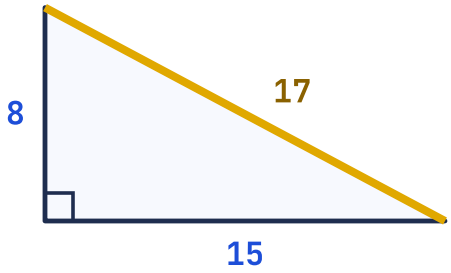
Watch out: The hypotenuse is **always** across from the square corner. Find it first.



$9 + 16 = 25$

$a^2 + b^2 = c^2$

Walk through: Which side is the hypotenuse?

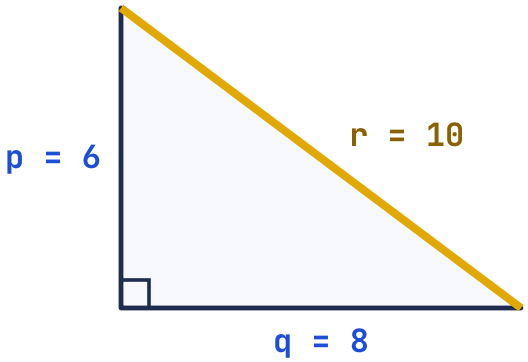


1. Where is the right angle (square corner)?
   Between the sides 8 and 15 → legs: 8 and 15
2. The hypotenuse is across from the right angle. Which side?
   17 → hypotenuse c = 17
3. Check with the theorem: $8^2 + 15^2 = ?$
   $64 + 225 = 289 = 17^2$ → $64 + 225 = 289 = 17^2$ ✓

Check your understanding

1 Score 3 NC.8.G.6

Which side is the **hypotenuse**?



(A) r = 10 (B) p = 6

(C) q = 8

2 Score 3 NC.8.G.6

A right triangle has legs a and b and hypotenuse c. Which equation is true?

(A) $a^2 \cdot b^2 = c^2$ (B) $a^2 + c^2 = b^2$

(C) $a^2 + b^2 = c^2$ (D) $a + b = c$

5.2 Find the hypotenuse NC.8.G.7

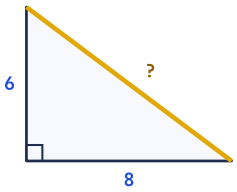
When the **hypotenuse is missing**: square each leg, **add**, then take the **square root**.

Square root (√)

undoes squaring: √100 = 10

Round to the nearest tenth

one digit after the decimal point: √13 ≈ 3.6



$6^2 + 8^2 = c^2$

$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

$6^2 + 8^2 = c^2$

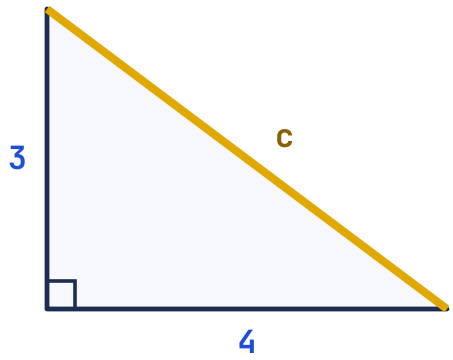
$36 + 64 = c^2$

$100 = c^2$

$c = \sqrt{100} = 10$

Watch out: Do not add the legs (6 + 8 = 14). Square them first.

Walk through: `Legs 3 and 4. Find the hypotenuse.`

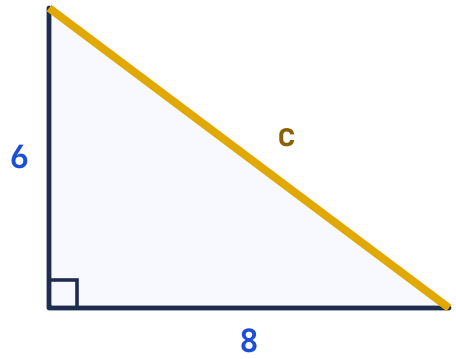


1. Write the theorem with the legs in place.
   $3^2 + 4^2 = c^2$ → $3^2 + 4^2 = c^2$
2. Square and add.
   $9 + 16 = 25$ → $c^2 = 25$
3. $c^2 = 25$. Find c.
   $c = \sqrt{25} = 5$ → $c = 5$

Check your understanding

1 Score 3 NC.8.G.7

The legs of a right triangle are **6** and **8**. Find the hypotenuse.

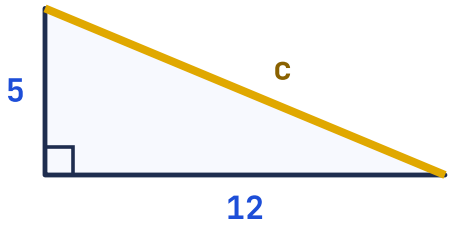


Answer: **10**

1. $6^2 + 8^2 = c^2$
2. $36 + 64 = c^2$
3. $c^2 = 100$
4. $c = \sqrt{100} = 10$

2 Score 3 NC.8.G.7

The legs of a right triangle are **5** and **12**. Find the hypotenuse.



Answer: **13**

1. $5^2 + 12^2 = c^2$
2. $25 + 144 = c^2$
3. $c^2 = 169$
4. $c = \sqrt{169} = 13$

5.3 NC.8.G.7

Find a missing leg

When a **leg is missing**: square the hypotenuse and the leg you know, **subtract**, then take the **square root**. `leg² = c² - (other leg)²`

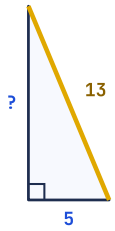
Hypotenuse → add

missing the long side: add the squares

Leg → subtract

missing a short side: subtract the squares

Watch out: If the missing side is a leg, the answer must be **shorter** than the hypotenuse. If it is not, you added by mistake.



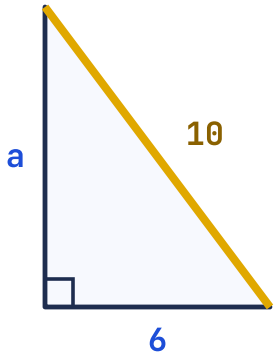
$a^2 + 5^2 = 13^2$

$a^2 + 25 = 169$

$a^2 = 169 - 25 = 144$

$a = \sqrt{144} = 12$

Walk through: `Hypotenuse 10, one leg 6. Find the other leg.`

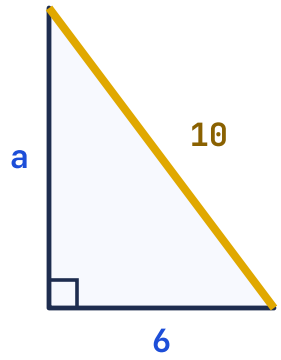


1. Which side is missing?
   A leg: subtract the squares → $a^2 + 6^2 = 10^2$
2. Square and subtract.
   $100 - 36 = 64 \rightarrow a^2 = 64$
3. Find a.
   $a = \sqrt{64} = 8 \rightarrow a = 8$

Check your understanding

1 Score 3 NC.8.G.7

A right triangle has hypotenuse **10** and one leg **6**. Find the other leg.

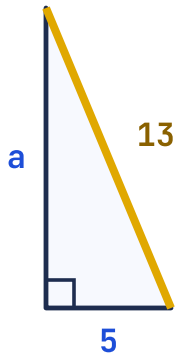


Answer: 8

1. $a^2 + 6^2 = 10^2$
2. $a^2 = 100 - 36 = 64$
3. $a = \sqrt{64} = 8$

2 Score 3 NC.8.G.7

A right triangle has hypotenuse **13** and one leg **5**. Find the other leg.



Answer: 12

1. $a^2 + 5^2 = 13^2$
2. $a^2 = 169 - 25 = 144$
3. $a = \sqrt{144} = 12$

5.4 Is it a right triangle? (the converse) NC.8.G.6

To test three side lengths: square the two **shorter** sides and add. If the sum **equals** the longest side squared, it **is** a right triangle. If not, it is not.

Converse
the theorem used backward, to test a triangle

Longest side
the only side that could be the hypotenuse

Watch out: Always use the **longest** side as c.

Sides 6, 8, 10 — $36 + 64 = 100$, $100 = 100$ ✓ — right triangle

Sides 4, 5, 6 — $16 + 25 = 41$, $41 \neq 36$ ✗ — not a right triangle

Walk through: Do 9, 12, 15 make a right triangle?

1. Which side would be the hypotenuse?
   15, the longest side → c = 15
2. Square and add the two shorter sides.
   81 + 144 = 225 → $9^2 + 12^2 = 225$

3 $15^2 = 225$. What do we decide?

Equal, so it IS a right triangle → 225 = 225 → right triangle

Check your understanding

1 Score 3 NC.8.G.6

Can the sides **6, 8, 10** make a right triangle?

- **A** **Yes: 36 + 64 = 100, which is 10^2**
- B No: the numbers are not 3, 4, 5
- C No: 6 + 8 is not 10

1. $6^2 + 8^2 = 36 + 64 = 100$
2. $10^2 = 100$
3. equal → right triangle

2 Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

- **A** **No: 16 + 25 = 41, but $6^2 = 36$**
- B Yes: 4 + 5 is more than 6
- C Yes: they are in counting order

1. $4^2 + 5^2 = 16 + 25 = 41$
2. $6^2 = 36$
3. not equal → not a right triangle

5.5 Distance on the coordinate plane NC.8.G.8

To find the distance between two points: draw a right triangle. The **across** change is one leg, the **up/down** change is the other leg, and the **distance** is the hypotenuse.

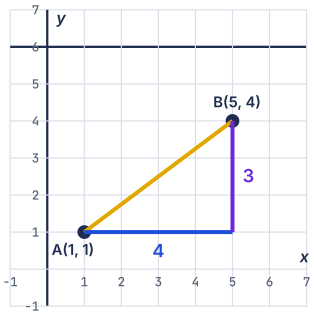
Run

the across change: subtract the x-values

Rise

the up/down change: subtract the y-values

Watch out: Count the length of each leg (always positive). Then use $a^2 + b^2 = c^2$.



run = 5 - 1 = 4

rise = 4 - 1 = 3

$4^2 + 3^2 = 16 + 9 = 25$

distance = √25 = 5

Walk through: Find the distance from (2, 1) to (8, 9).

1 Find the run (across change).

8 − 2 = 6 → run = 6

2 Find the rise (up/down change).

9 − 1 = 8 → rise = 8

3 Use $a^2 + b^2 = c^2$.

36 + 64 = 100, so the distance is 10 → distance = √100 = 10

Check your understanding

1 Score 3 NC.8.G.8

Find the distance from **(1, 2)** to **(4, 6)**.

Answer: **5**

1. `run = 4 - 1 = 3, rise = 6 - 2 = 4`
2. $3^2 + 4^2 = 9 + 16 = 25$
3. `distance = √25 = 5`

SKILL BUILDERS · UNIT 5 · SCORE 3

Pythagorean Theorem: Practice Set

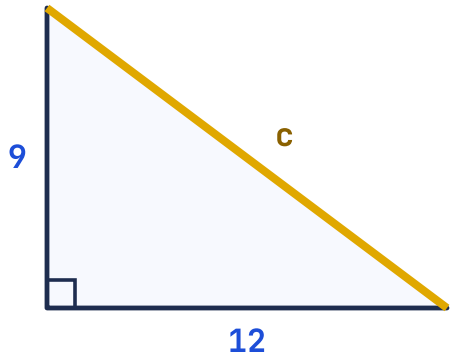
NC.8.G.6–8

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.7

The legs of a right triangle are **9** and **12**. Find the hypotenuse.

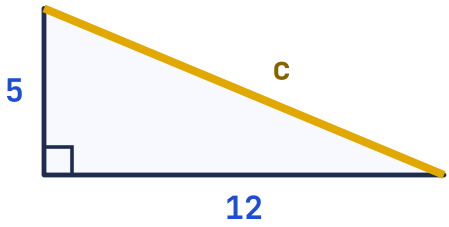


Answer: **15**

1. $9^2 + 12^2 = c^2$
2. $81 + 144 = c^2$
3. $c^2 = 225$
4. $c = \sqrt{225} = 15$

2 Score 3 NC.8.G.7

The legs of a right triangle are **5** and **12**. Find the hypotenuse.

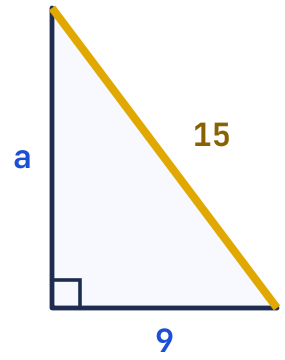


Answer: **13**

1. $5^2 + 12^2 = c^2$
2. $25 + 144 = c^2$
3. $c^2 = 169$
4. $c = \sqrt{169} = 13$

3 Score 3 NC.8.G.7

A right triangle has hypotenuse **15** and one leg **9**. Find the other leg.

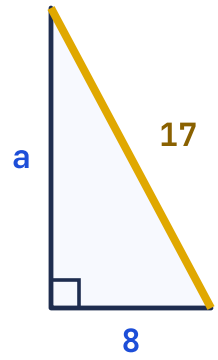


Answer: **12**

1. $a^2 + 9^2 = 15^2$
2. $a^2 = 225 - 81 = 144$
3. $a = \sqrt{144} = 12$

4 Score 3 NC.8.G.7

A right triangle has hypotenuse **17** and one leg **8**. Find the other leg.



Answer: **15**

1. $a^2 + 8^2 = 17^2$
2. $a^2 = 289 - 64 = 225$
3. $a = \sqrt{225} = 15$

5 Score 3 NC.8.G.6

Can the sides **7, 24, 25** make a right triangle?

(A) Yes: $49 + 576 = 625$, which is 25^2

(B) No: the numbers are not 3, 4, 5

(C) No: $7 + 24$ is not 25

1. $7^2 + 24^2 = 49 + 576 = 625$
2. $25^2 = 625$
3. equal → right triangle

6 Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

(A) No: $16 + 25 = 41$, but $6^2 = 36$

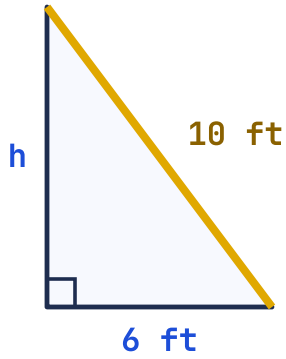
(B) Yes: $4 + 5$ is more than 6

(C) Yes: they are in counting order

1. $4^2 + 5^2 = 16 + 25 = 41$
2. $6^2 = 36$
3. not equal → not a right triangle

7 Score 3 NC.8.G.7

A 10-foot ladder leans on a wall. The bottom is **6 feet** from the wall. How high up the wall does it reach?

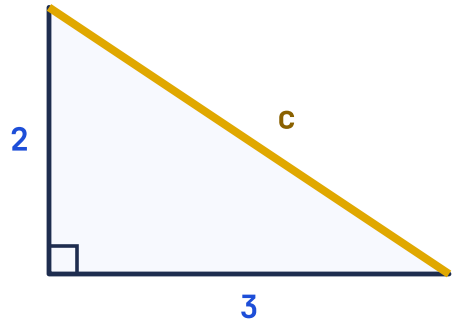


Answer: **8ft**

1. $a^2 + 6^2 = 10^2$
2. $a^2 = 100 - 36 = 64$
3. $a = \sqrt{64} = 8$

8 Score 3 NC.8.G.7

The legs of a right triangle are **2** and **3**. Find the hypotenuse to the nearest tenth.



Answer: **3.6**

1. $2^2 + 3^2 = c^2$
2. $4 + 9 = c^2$
3. $c^2 = 13$
4. $c = \sqrt{13} \approx 3.6$

9 Score 3 NC.8.G.8

Find the distance from **(−2, 1)** to **(2, 4)**.

Answer: **5**

1. run = 2 - (-2) = 4, rise = 4 - 1 = 3
2. $4^2 + 3^2 = 16 + 9 = 25$
3. distance = $\sqrt{25} = 5$

10 Score 3 NC.8.G.7

Find the mistake. A right triangle has legs 6 and 8. Max wrote: $6 + 8 = 14$, so $c = 14$. What went wrong?

(A) Max is right: c = 14.

(B) Max added the legs. He must square them: 36 + 64 = 100, so c = 10.

(C) Max should subtract: 8 − 6 = 2.

(D) Max should multiply: 6 × 8 = 48.

Functions: Mini-Lessons

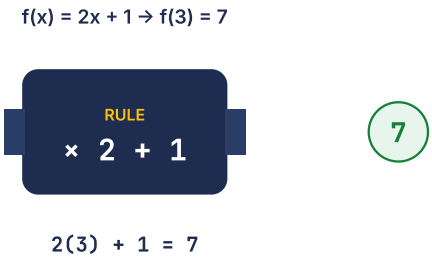
6.1 What is a function? NC.8.F.1

A **function** is a rule where **each input (x) has exactly one output (y)**. f(x) means "the output when the input is x." To find f(4), put 4 in for x.

Input (x)
the number you put in

Output (y or f(x))
the number that comes out

Function
every input has exactly one output



Not a function: one input with **two** outputs, like x = 2 going to both 5 and 7. On a graph, a vertical line that hits the graph twice means it is **not** a function.

Watch out: Two inputs with the same output is still a function. The problem is one input with two outputs.

Walk through: `f(x) = 3x - 2. Find f(4).`

1. What does f(4) ask?
   Put 4 in for x → f(4) = 3(4) − 2
2. Multiply first: 3(4) = ?
   12 → = 12 − 2
3. Finish: 12 − 2 = ?
   10 → f(4) = 10

Check your understanding

1 Score 3 NC.8.F.1

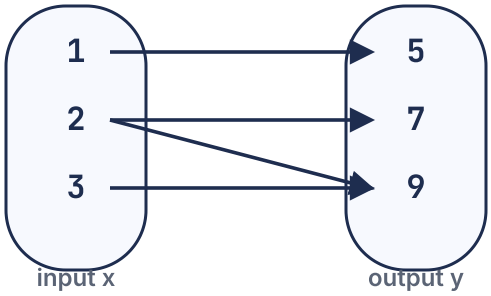
`f(x) = 2x + 5`. Find `f(3)`.

Answer: **11**

1. `f(3) = 2(3) + 5`
2. `= 6 + 5 = 11`

2 Score 3 NC.8.F.1

Is this mapping a function?



A **No: the input 2 has two outputs**

Ⓑ Yes: every input has an arrow

Ⓒ Yes: two inputs share an output

6.2 Slope: rise over run NC.8.F.4

Slope tells how steep a line is. **`slope = rise ÷ run = (change in y) ÷ (change in x)`**. Uphill (left to right) is **positive**, downhill is **negative**, flat is **0**.

Rise
the up/down change: subtract the y-values

Run
the across change: subtract the x-values

Slope (m)
rise ÷ run



`slope = rise ÷ run`

`= (8 - 2) ÷ (3 - 0)`

`= 6 ÷ 3 = 2`

up 2 for every 1 across

Green: positive (uphill) · Red: negative (downhill) · Gray: zero (flat)

Watch out: Rise goes on **top**. Run goes on the bottom. And subtract in the same order for both.

Walk through: `Find the slope from (1, 3) to (4, 12).`

1. Find the rise (change in y).
   12 − 3 = 9 → rise = 9
2. Find the run (change in x).
   4 − 1 = 3 → run = 3

3 Slope = rise ÷ run = ?

9 ÷ 3 = 3 → slope = 3

Check your understanding

1 Score 3 NC.8.F.4

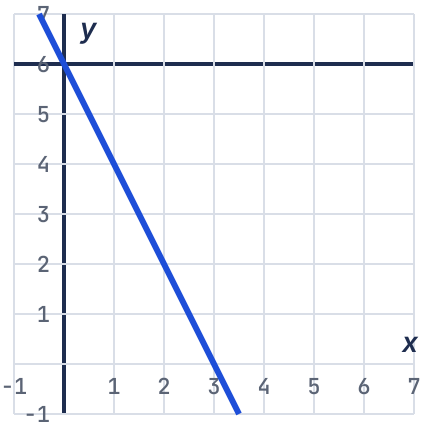
Find the slope of the line through **(0, 2)** and **(3, 8)**.

Answer: 2

1. `rise = 8 - 2 = 6`
2. `run = 3 - 0 = 3`
3. `slope = 6 ÷ 3 = 2`

2 Score 3 NC.8.F.4

What kind of slope does this line have?



(A) **Negative** (B) Positive

(C) Zero

6.3 y = mx + b from a table or graph NC.8.F.4

A line can be written **y = mx + b**. **m** is the slope (how much y changes when x goes up 1). **b** is the y-intercept (the y-value when x = 0, where the line crosses the y-axis).

Slope-intercept form

y = mx + b

y-intercept (b)

where the line crosses the y-axis; the y-value when x = 0

x	0	1	2	3
y	5	8	11	14

y goes up 3 each time → m = 3. When x = 0, y = 5 → b = 5. So y = 3x + 5.

Watch out: In y = 4x − 1, the slope is 4 and the y-intercept is **−1** (keep the sign).

x	0	1	2	3
y	2	5	8	11

+3 +3 +3

`m = 3`

`b = 2`

`y = 3x + 2`

m = slope (rate) · b = y-intercept (start)

Walk through: `Write the equation for this table.`

1

x	0	1	2	3
y	1	3	5	7

How much does y change when x goes up 1?

+2, so m = 2 → m = 2

2 What is y when x = 0?

1, so b = 1 → b = 1

3 Write y = mx + b.

y = 2x + 1 → y = 2x + 1

Check your understanding

1 Score 3 NC.8.F.3

In **y = 4x - 1**, what is the **slope**?

Answer: **4**

2 Score 3 NC.8.F.4

Which equation matches the table?

x	0	1	2	3
y	4	6	8	10

(A) **y = 2x + 4**

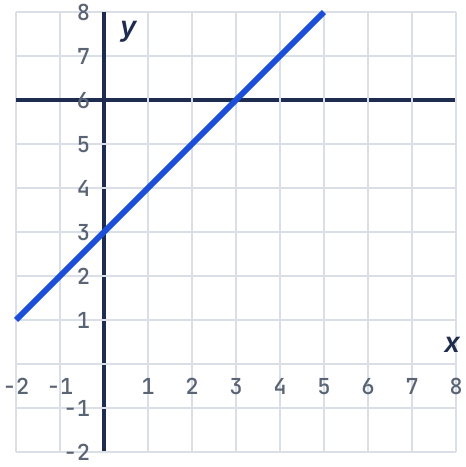
(B) y = 2x

(C) y = 4x + 2

(D) y = x + 4

3 Score 3 NC.8.F.4

What is the y-intercept of this line?



Answer: **3**

6.4 Rate and start value NC.8.F.2

In a real-world line, the **slope is the rate** (per hour, per mile, each day) and the **y-intercept is the start value** (the amount when x = 0, like a fee).

Rate of change
how much y changes for each 1 of x

Initial (start) value
the amount at the start, when x = 0

Increasing / decreasing
a graph going up / going down from left to right



`Taxi: y = 2x + 3`

2 = rate: $2 per mile

3 = start: $3 before driving

`5 miles: 2(5) + 3 = $13`

Phone plan: **`y = 15x + 40`**. $40 to start, then $15 each month.

Watch out: The start value is the number **by itself**, not the number next to x.

Walk through: `A gym charges y = 20x + 50 for x months. What does 50 mean?`

1. In y = 20x + 50, which number is alone (not next to x)?
   50 → b = 50
2. What does the number alone mean?
   The start fee: $50 before any months → $50 is the start (sign-up) fee
3. And 20?
   $20 per month (the rate) → $20 per month is the rate

Check your understanding

1 Score 3 NC.8.F.2

A plant is **`h = 2w + 6`** cm tall after w weeks. How tall was it at the start?

Answer: **6cm**

2 Score 3 NC.8.F.2

A phone plan costs **`y = 15x + 40`**. What is the cost **per month**?

Answer: **15dollars**

3 Score 3 NC.8.F.5

A graph of water in a tank goes **down** from left to right. What is happening?

Ⓐ **The water is decreasing**

Ⓑ The water is increasing

Ⓒ The water stays the same

6.5 Systems: where two lines cross NC.8.EE.8

The **solution of a system** is the point **(x, y) where the two lines cross**. It makes **both** equations true.

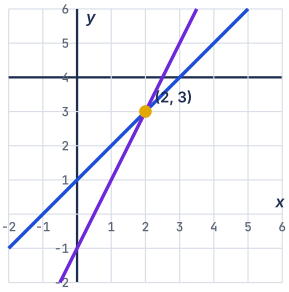
System of equations

two equations used together

Solution

the point that is on both lines

Watch out: A point on only one line is not the solution. Check it in **both** equations.



$y = x + 1$

$y = 2x - 1$

Check (2, 3):

$3 = 2 + 1$ ✓

$3 = 2(2) - 1$ ✓

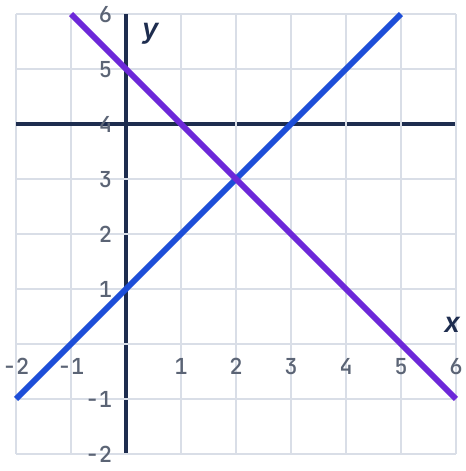
Walk through: Is (1, 3) the solution of y = x + 2 and y = 3x?

1. Check the first equation: y = x + 2 with x = 1.
   1 + 2 = 3 ✓ (y is 3) → y = x + 2: 3 = 1 + 2 ✓
2. Check the second equation: y = 3x with x = 1.
   3(1) = 3 ✓ → y = 3x: 3 = 3(1) ✓
3. Is (1, 3) the solution?
   Yes: it works in both → (1, 3) is the solution

Check your understanding

1 Score 3 NC.8.EE.8

What is the solution of the system shown?



Ⓐ (0, 1)

Ⓑ **(2, 3)**

Ⓒ (3, 2)

Ⓓ (0, 5)

2 Score 3 NC.8.EE.8

Is **(2, 5)** the solution of **y = x + 3** and **y = 2x**?

Ⓐ **No: it works in y = x + 3, but 2(2) = 4, not 5**

Ⓑ Yes: 2 + 3 = 5

Ⓒ Yes: it works in both

SKILL BUILDERS · UNIT 6 · SCORE 3

Functions: Practice Set

NC.8.F.1–5 · NC.8.EE.8

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.F.1

$f(x) = 4x + 1$. Find $f(2)$.

Answer: 9

1. $f(2) = 4(2) + 1$
2. $= 8 + 1 = 9$

2 Score 3 NC.8.F.1

Which table shows a function?

(A)

x	1	1	2	3
y	4	5	6	7

(B)

x	2	3	2	4
y	1	2	3	4

(C)

x	1	2	3	4
y	5	5	6	6

3 Score 3 NC.8.F.4

Find the slope from **(1, 3)** to **(4, 12)**.

Answer: 3

1. rise = 12 - 3 = 9
2. run = 4 - 1 = 3
3. slope = 9 ÷ 3 = 3

4 Score 3 NC.8.F.4

Find the slope from **(0, 6)** to **(3, 0)**.

Answer: -2

1. rise = 0 - 6 = -6
2. run = 3 - 0 = 3
3. slope = -6 ÷ 3 = -2

5 Score 3 NC.8.F.4

Which equation matches the table?

x	0	1	2
y	5	8	11

Ⓐ y = 3x

Ⓑ **y = 3x + 5**

Ⓒ y = 5x + 3

Ⓓ y = x + 5

1. `m = 3 (y goes up 3 each step)`
2. `b = 5 (y when x = 0)`
3. `y = 3x + 5`

6 Score 3 NC.8.F.2

A savings account has `y = 5x + 20` dollars after x weeks. What is the **start value**?

Answer: **20dollars**

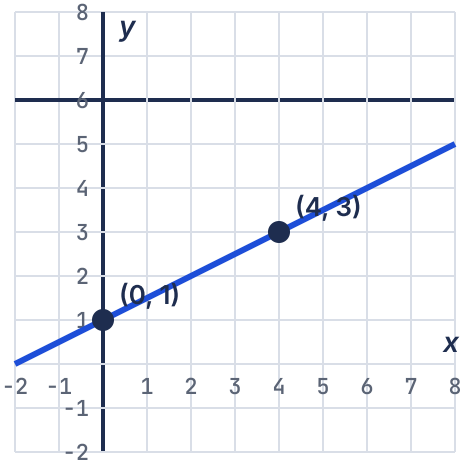
7 Score 3 NC.8.F.2

In `y = 5x + 20`, what is the **rate** (amount added each week)?

Answer: **5dollars**

8 Score 3 NC.8.F.4

What is the slope of this line?



Answer: **0.5**

1. `rise = 3 - 1 = 2`
2. `run = 4 - 0 = 4`
3. `slope = 2 ÷ 4 = 1/2`

9 Score 3 NC.8.EE.8

Solve the system: **y = x + 2** and **y = 3x**.

A **(1, 3)**

Ⓑ (2, 4)

Ⓒ (3, 1)

Ⓓ (0, 2)

1 $x + 2 = 3x$ ↓ Subtract x

$-x \quad -x$

2 $\frac{2}{2} = \frac{2x}{2}$ ↓ Divide by 2

3 $1 = x$

1 y = 3(1) = 3 → (1, 3)

10 Score 3 NC.8.F.5

A runner's distance-from-home graph is **flat** for a while. What is the runner doing?

A **Stopped: the distance is not changing**

Ⓑ Running faster

Ⓒ Running home

11 Score 3 NC.8.F.4

Find the mistake. For (2, 1) and (4, 7), Kai wrote: `slope = (4 - 2) ÷ (7 - 1) = 2 ÷ 6 = 1/3`. What went wrong?

A **Kai put the run on top. Slope = rise ÷ run = 6 ÷ 2 = 3.**

Ⓑ Kai is right.

Ⓒ The slope should be negative.

Ⓓ Kai should add: (4 + 2) ÷ (7 + 1).

Statistics: Mini-Lessons

7.1 Scatter plots and correlation NC.8.SP.1

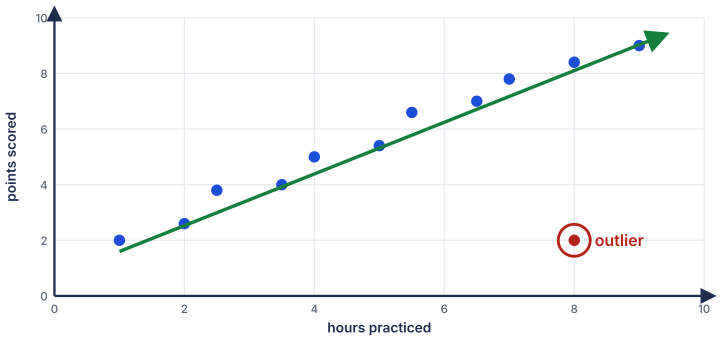
A **scatter plot** shows two sets of data as dots. Describe it: **direction** (positive = up, negative = down, none), **strength** (close to a line = strong), and **outliers** (dots far from the rest).

Positive correlation
as x goes up, y goes up

Negative correlation
as x goes up, y goes down

Outlier
a dot far away from the pattern

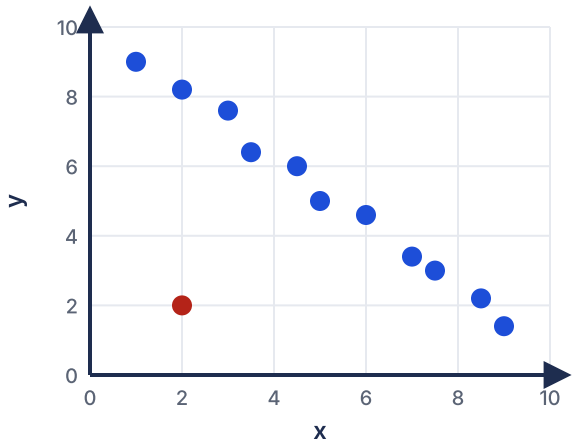
Correlation is not causation
two things moving together does not prove one causes the other



Left: negative (down). Right: no correlation (no pattern).

Watch out: Ice cream sales and sunburns both go up in summer. Ice cream does not **cause** sunburns: hot weather drives both.

Walk through: `Describe this scatter plot.`



1 Read left to right. Which way do most dots go?
Down: negative correlation → direction: negative

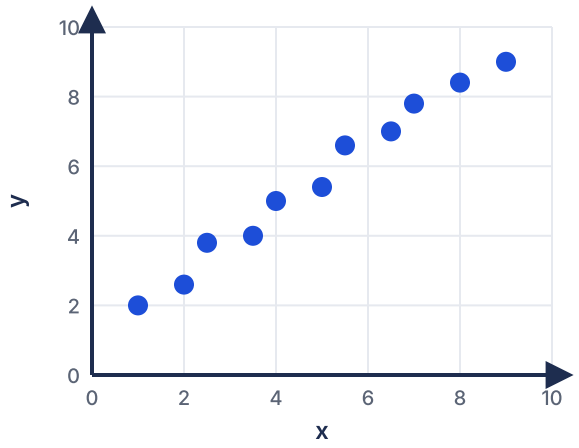
2 Are the dots close to a line or spread out?
Close to a line: strong → strength: strong

3 Is there an outlier?
Yes: the dot at (2, 2) → outlier: (2, 2)

Check your understanding

1 Score 3 NC.8.SP.1

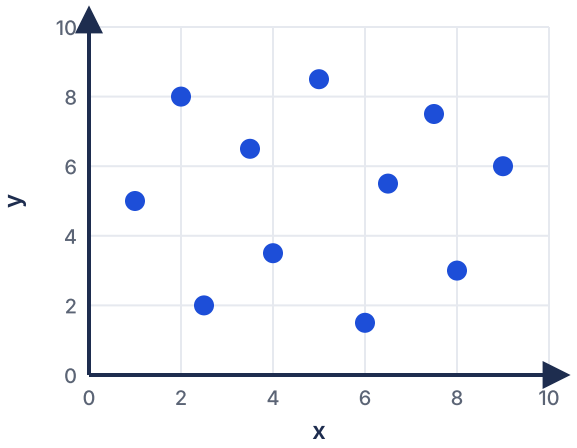
What kind of correlation does this scatter plot show?



(A) **Positive correlation**
(B) Negative correlation
(C) No correlation

2 Score 3 NC.8.SP.1

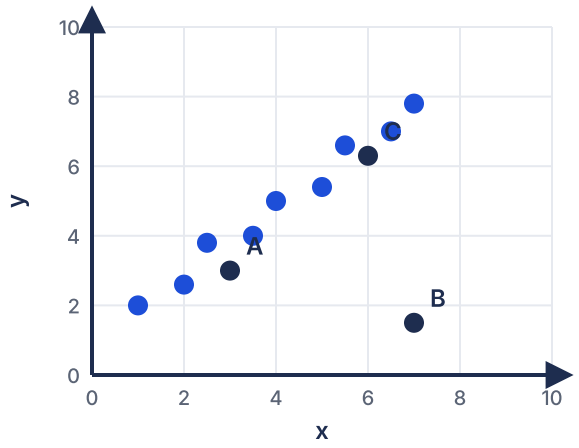
What kind of correlation does this scatter plot show?



(A) **No correlation**
(B) Positive correlation
(C) Negative correlation

3 Score 3 NC.8.SP.1

Which point is an **outlier**?



(A) **B**
(B) A
(C) C

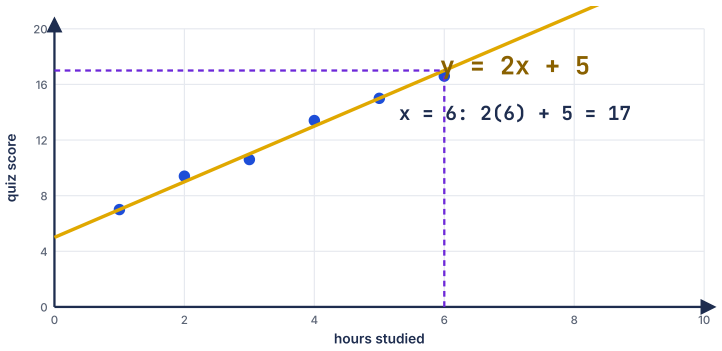
7.2 Line of best fit and predictions NC.8.SP.2

A **line of best fit** is a straight line through the middle of the dots. It is just **y = mx + b**: plug in an x to **predict** y.

Line of best fit
a line close to most dots, about as many above as below

Interpolation
predicting inside the data (safer)

Extrapolation
predicting outside the data (riskier)



Watch out: A good fit line does not have to touch any dot. It goes through the middle of the group.

Walk through:

Best fit: y = 3x + 4 (x = hours, y = score). Predict y for x = 5.

1. What do we do with x = 5?
   Put 5 in for x: y = 3(5) + 4 → y = 3(5) + 4
2. 3(5) = ?
   15 → y = 15 + 4
3. Finish.
   19 → y = 19

Check your understanding

1 Score 3 NC.8.SP.3

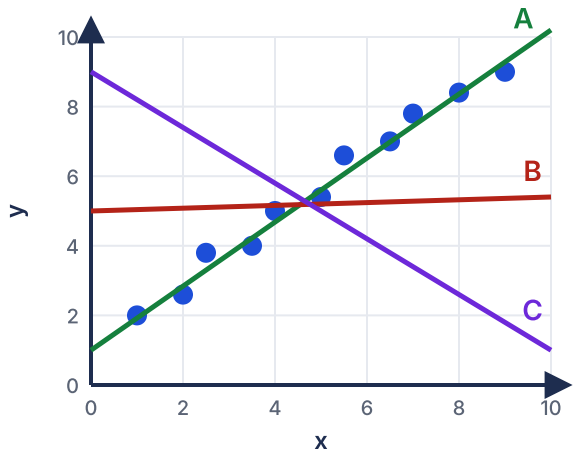
Best fit: $y = 2x + 5$. Predict y when x = 6.

Answer: 17

1. $y = 2(6) + 5$
2. $= 12 + 5 = 17$

2 Score 3 NC.8.SP.2

Which line fits the data best?



(A) Line A

(B) Line B

(C) Line C

7.3 Two-way tables NC.8.SP.4

A **two-way table** counts data sorted two ways. Inside boxes are **joint** counts; the edge totals are **marginal** counts. **Relative frequency** = count ÷ total, often as a percent.

Joint frequency

one inside box

Marginal frequency

a row or column total

Relative frequency

a count divided by a total: 12 ÷ 40 = 30%

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

	Likes dogs	Does not	Total
Boys	12	8	20
Girls	15	5	20
Total	27	13	40

12 ÷ 20 = 0.60 = 60%

Watch out: Always ask "out of which total?" Out of everyone (40), or out of one row (20)?

Walk through:

Use the table. What percent of all students are girls who like dogs?

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

1. Find the box: girls who like dogs.
   15 → girls who like dogs: 15
2. Out of which total?
   All students: 40 → 15 ÷ 40
3. 15 ÷ 40 = ?
   0.375 = 37.5% → 15 ÷ 40 = 0.375 = 37.5%

Check your understanding

1 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

How many **boys** are there in all?

Answer: **20**

2 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

What percent of **all** students do **not** like dogs?

Answer: **32.5%**

1. `does not like dogs: 13`
2. `13 ÷ 40 = 0.325 = 32.5%`

SKILL BUILDERS · UNIT 7 · SCORE 3

Statistics: Practice Set

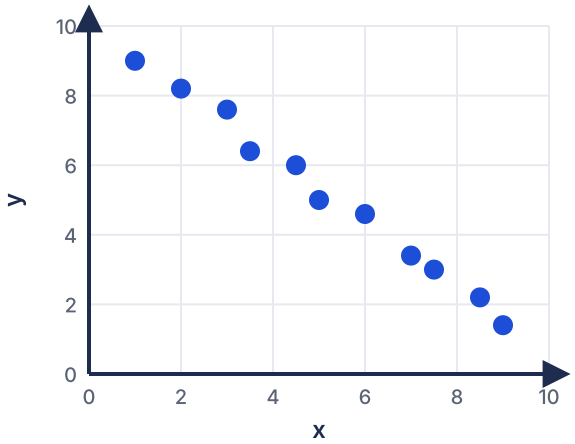
NC.8.SP.1–4

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



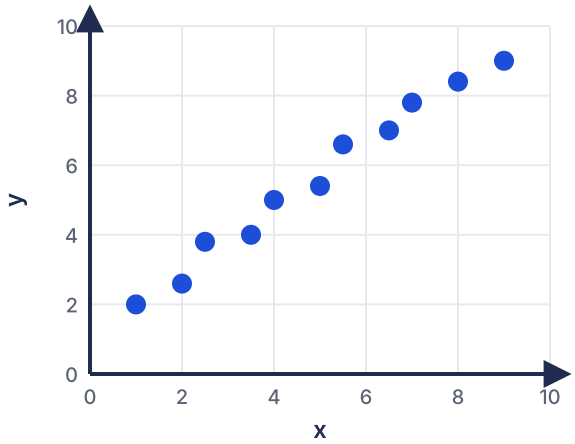
A **Negative correlation**

B Positive correlation

C No correlation

2 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



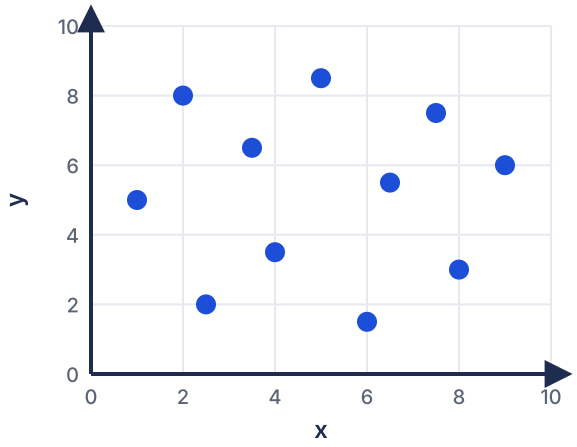
A **Positive correlation**

B Negative correlation

C No correlation

3 Score 3 NC.8.SP.1

What kind of correlation does this scatter plot show?



A **No correlation**

B Positive correlation

C Negative correlation

4 Score 3 NC.8.SP.1

A single dot far away from the rest of a scatter plot is called...

A **an outlier**

B a line of best fit

C a correlation

5 Score 3 NC.8.SP.3

Best fit: **y = 3x + 4**. Predict y when x = 5.

Answer: **19**

1 `y = 3(5) + 4`

2 `= 15 + 4 = 19`

6 Score 3 NC.8.SP.3

A best fit line for plant height is **y = 2x + 3** (x = weeks, y = cm). What does the slope 2 mean?

A The plant is 2 weeks old

B **The plant grows about 2 cm each week**

C The plant started at 2 cm

7 Score 3 NC.8.SP.2

Data go from x = 1 to x = 8. Using the line to predict y at x = 30 is...

A **extrapolation (riskier)**

B interpolation (safer)

8 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

How many **girls** like dogs?

Answer: **15**

9 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

Of the **boys**, what percent like dogs?

Answer: **60%**

1 `boys who like dogs: 12; all boys: 20`

2 `12 ÷ 20 = 0.60 = 60%`

10 Score 3 NC.8.SP.1

Find the mistake. A scatter plot shows that towns with more ice cream shops have more sunburns. Rio says: "Ice cream causes sunburns." What is wrong?

(A) Nothing. The data prove it.

(B) A correlation does not prove cause. Hot, sunny weather can cause both.

(C) The correlation must be negative.

(D) Sunburns cause ice cream shops.

Transformations: Mini-Lessons

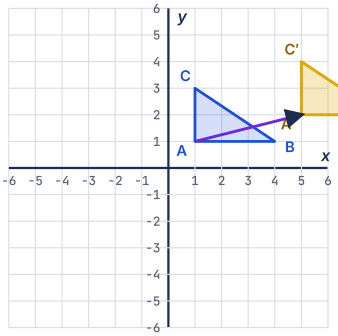
8.1 Translations (slides) NC.8.G.3

A **translation** slides a figure. **Right/left changes x** (add or subtract). **Up/down changes y**. The image is **congruent** (same size and shape).

Pre-image
the original figure

Image
the figure after the move, named with ′ (A′, "A prime")

Congruent
same size and same shape



A(1, 1) → A′(5, 2)
B(4, 1) → B′(8, 2)
C(1, 3) → C′(5, 4)

(x, y) → (x + 4, y + 1)

same size and shape: congruent

Move	right h	left h	up k	down k
Do	x + h	x − h	y + k	y − k

Watch out: Left and right change **x** (the first number). Up and down change **y** (the second number).

Walk through: `Translate (1, -2) left 3 and up 5.`

1. Left 3 changes which number?
   x: 1 − 3 = −2 → x: 1 − 3 = −2
2. Up 5 changes which number?
   y: −2 + 5 = 3 → y: −2 + 5 = 3
3. The image point is…
   (−2, 3) → (1, −2) → (−2, 3)

Check your understanding

1 Score 3 NC.8.G.3

Translate right 4 and up 1: the point **(2, 3)**. Where does it land?

Ⓐ (−2, 4) Ⓑ (3, 7)

Ⓒ **(6, 4)** Ⓓ (6, 2)

2 Score 3 NC.8.G.3

Translate left 2 and down 3: the point **(2, 5)**. Where does it land?

Ⓐ (0, 8) Ⓑ (−1, 3)

Ⓒ (4, 8) Ⓓ **(0, 2)**

8.2 Reflections (flips) NC.8.G.3

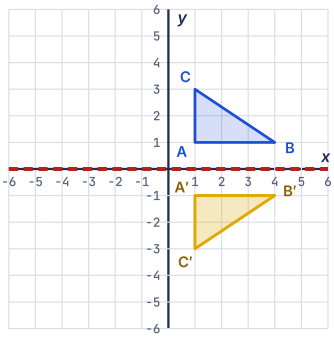
A **reflection** flips a figure over a line. Over the **x-axis**: y changes sign, `(x, y) → (x, -y)`. Over the **y-axis**: x changes sign, `(x, y) → (-x, y)`.

Line of reflection

the mirror line

Opposite

same number, other sign: the opposite of 5 is −5



`A(1, 1) → A'(1, -1)`
`B(4, 1) → B'(4, -1)`
`C(1, 3) → C'(1, -3)`

`(x, y) → (x, -y)`

over the x-axis: y changes sign

Trick: you flip **over** the x-axis, so the **other** letter (y) changes sign.

Watch out: Only **one** coordinate changes sign in a reflection over an axis.

Walk through: `Reflect (3, 5) over the y-axis.`

1 Over the y-axis, which coordinate changes sign?
x → x changes sign

2 The image is...
(−3, 5) → (3, 5) → (−3, 5)

Check your understanding

1 Score 3 NC.8.G.3

Reflect over the x-axis: the point **(3, 5)**. Where does it land?

Ⓐ **(3, −5)** Ⓑ (−3, 5)

Ⓒ (5, 3) Ⓓ (−3, −5)

2 Score 3 NC.8.G.3

Reflect over the y-axis: the point **(−3, 4)**. Where does it land?

Ⓐ **(3, 4)** Ⓑ (4, −3)

Ⓒ (3, −4) Ⓓ (−3, −4)

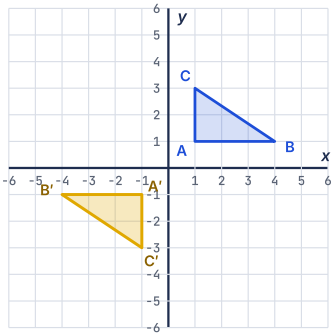
8.3 Rotations (turns) NC.8.G.3

A **rotation** turns a figure around the origin. **180°**: `(x, y) → (-x, -y)`. **90° counterclockwise**: `(x, y) → (-y, x)`. **270° counterclockwise** (90° clockwise): `(x, y) → (y, -x)`.

Counterclockwise
the opposite way a clock's hands turn

Origin
the point (0, 0) where the axes cross

Turn	Rule	(4, 2) →
90° CCW	(−y, x)	(−2, 4)
180°	(−x, −y)	(−4, −2)
270° CCW	(y, −x)	(2, −4)



`A(1, 1) → A'(-1, -1)`
`B(4, 1) → B'(-4, -1)`
`C(1, 3) → C'(-1, -3)`

`(x, y) → (-x, -y)`
180° turn: both signs change

Watch out: For 90° and 270°, the numbers **switch places**. For 180°, they do not.

Walk through: `Rotate (1, 5) 90° counterclockwise.`

1. Which rule is 90° counterclockwise?
   (x, y) → (−y, x) → (x, y) → (−y, x)
2. Here x = 1 and y = 5. What is −y?
   −5 → −y = −5
3. The image is…
   (−5, 1) → (1, 5) → (−5, 1)

Check your understanding

1 Score 3 NC.8.G.3

Rotate 180° about the origin: the point **(4, 2)**. Where does it land?

Ⓐ (2, 4) Ⓑ **(−4, −2)**
Ⓒ (−4, 2) Ⓓ (−2, 4)

2 Score 3 NC.8.G.3

Rotate 90° counterclockwise about the origin: the point **(5, 5)**. Where does it land?

Ⓐ (5, 5) Ⓑ (−5, −5)
Ⓒ (5, −5) Ⓓ **(−5, 5)**

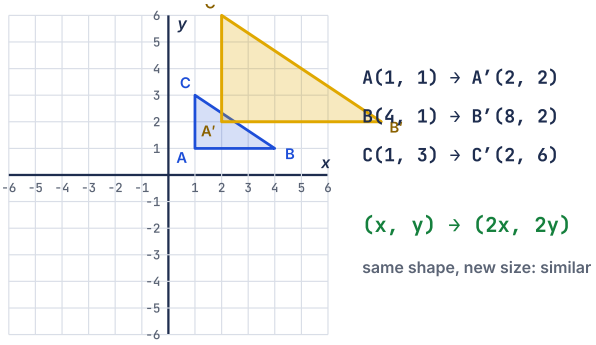
8.4 Dilations, congruent and similar NC.8.G.4

A **dilation** multiplies both coordinates by the **scale factor k**: `(x, y) → (kx, ky)`. k > 1 makes it bigger; k between 0 and 1 makes it smaller. The image is **similar**. Slides, flips and turns keep the image **congruent**.

Scale factor (k)
the number you multiply by

Similar
same shape, maybe a different size

Congruent
same shape AND same size



Dilation = resize = similar. Translation, reflection, rotation = same size = congruent.

Watch out: Multiply **both** numbers by k. Do not add k.

Walk through: `Dilate (3, -4) by k = 3.`

1. What do we do with k = 3?
   Multiply both coordinates by 3 → (3 · 3, −4 · 3)
2. Multiply.
   (9, −12) → (3, −4) → (9, −12)
3. Is the image congruent or similar?
   Similar: same shape, bigger size → similar (not congruent)

Check your understanding

1 Score 3 NC.8.G.4

Dilate by k = 2: the point **(2, 3)**. Where does it land?

- **(A) (4, 6)**
- (B) (4, 5)
- (C) (1, 1.5)
- (D) (2, 6)

2 Score 3 NC.8.G.4

After a **dilation** with k = 3, the image is...

- **(A) similar to the pre-image**
- (B) congruent to the pre-image
- (C) neither congruent nor similar

3 Score 3 NC.8.G.2

After a **reflection**, the image is...

A congruent to the pre-image

B bigger than the pre-image

C smaller than the pre-image

SKILL BUILDERS · UNIT 8 · SCORE 3

Transformations: Practice Set

NC.8.G.2–4

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.3

Translate left 2 and down 3: the point **(2, 5)**. Where does it land?

- (A) (−1, 3)
- **(B) (0, 2)**
- (C) (4, 8)
- (D) (0, 8)

2 Score 3 NC.8.G.3

Reflect over the y-axis: the point **(−3, 4)**. Where does it land?

- (A) (4, −3)
- (B) (−3, −4)
- (C) (3, −4)
- **(D) (3, 4)**

3 Score 3 NC.8.G.3

Reflect over the x-axis: the point **(6, −1)**. Where does it land?

- **(A) (6, 1)**
- (B) (−6, 1)
- (C) (−1, 6)
- (D) (−6, −1)

4 Score 3 NC.8.G.3

Rotate 180° about the origin: the point **(2, 7)**. Where does it land?

- **(A) (−2, −7)**
- (B) (−2, 7)
- (C) (−7, 2)
- (D) (7, 2)

5 Score 3 NC.8.G.4

Dilate by k = 4: the point **(1, 3)**. Where does it land?

- (A) (4, 3)
- **(B) (4, 12)**
- (C) (5, 7)
- (D) (0.25, 0.75)

6 Score 3 NC.8.G.3

Rotate 90° counterclockwise about the origin: the point **(5, 5)**. Where does it land?

- (A) (−5, −5)
- (B) (5, −5)
- (C) (5, 5)
- **(D) (−5, 5)**

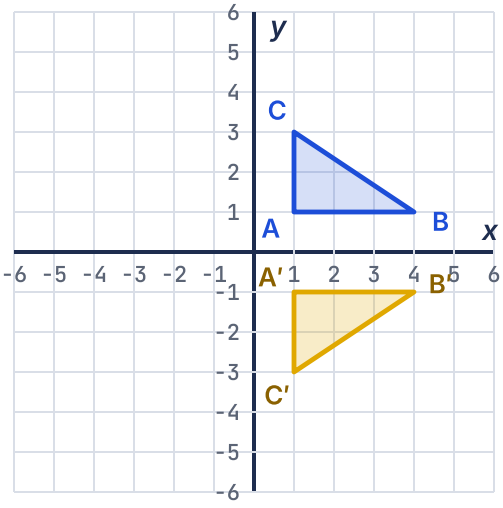
7 Score 3 NC.8.G.4

Dilate by k = 1/2: the point **(4, −6)**. Where does it land?

- Ⓐ (4.5, −5.5)
- Ⓑ **(2, −3)**
- Ⓒ (2, 3)
- Ⓓ (8, −12)

8 Score 3 NC.8.G.3

Which move takes the blue triangle to the gold triangle?



- Ⓐ Translation down 4
- Ⓑ **Reflection over the x-axis**
- Ⓒ Dilation by k = 2
- Ⓓ Rotation 180°

9 Score 3 NC.8.G.2

After a translation, a reflection and a rotation, the image is...

- Ⓐ **congruent: same size and shape**
- Ⓑ similar but smaller
- Ⓒ similar but bigger

10 Score 3 NC.8.G.3

First translate right 3, then reflect over the x-axis: the point **(1, 2)**. Where does it land?

- Ⓐ (4, 2)
- Ⓑ **(4, −2)**
- Ⓒ (−4, −2)
- Ⓓ (1, −5)

1 `(1, 2) → right 3 → (4, 2)`

2 `(4, 2) → over x-axis → (4, -2)`

11 Score 3 NC.8.G.3

Find the mistake. Mia reflected (2, 6) over the x-axis and got `(-2, 6)`. What went wrong?

A Mia changed x. Over the x-axis, y changes sign: (2, −6).

B Mia is right.

C The numbers should switch: (6, 2).

D Both signs should change: (−2, −6).